# A CNN-Transformer Hybrid Network With Boundary Guidance for Mapping Cropland Field Parcels From High-Resolution Remote Sensing Imagery

Junyang Xie, Hao Wu, *Senior Member, IEEE*, Wenbin Wu, Liang Hong, Lihua He, Qiangyi Yu, Lanfa Liu, Anqi Lin, and Jaturong Som-ard

***Abstract*—Accurate mapping of cropland field parcels (CFPs) is essential for efficient agricultural production and management. However, CFPs in high-resolution remote sensing imagery exhibit substantial variability in size, shape, and spatial distribution, making it challenging to accurately capture their features using only local or global information, thereby limiting the mapping precision. To address this, we propose a convolutional neural network (CNN)-transformer hybrid network with boundary guidance (CTHBNet) for effective CFP mapping from high-resolution imagery. CTHBNet comprises three key modules: 1) a CNN-transformer fusion encoder that integrates local and global features to enhance recognition of multiscale parcels; 2) a hierarchical information fusion decoder that progressively restores parcel shapes to delineate complete boundaries; and 3) a boundary-guided feature enhancement module (BGFE-Module) that refines the boundary clarity and reduces parcel adhesion. Moreover, a multitask learning (MTL) strategy jointly optimizes parcel extent, boundary, and distance features to improve the mapping precision. We evaluated CTHBNet on GaoFen-2 (GF-2) imagery across four distinct agricultural regions in China. The results showed that CTHBNet can achieve an overall accuracy (OA) of over 92%, an F1-score exceeding 82%, and a mean intersection over union (mIoU) exceeding 84%, demonstrating its effectiveness in CFP mapping. Comparative experiments confirmed the superior accuracy and boundary delineation of CTHBNet, and ablation experiments validated the effectiveness of each proposed module. Furthermore, experiments on five global regions and multiple public benchmarks further demonstrated its robust generalization capability. In particular, by fusing the complementary strengths of CNN and transformer models, CTHBNet significantly improves the representation of complex parcel boundaries and diverse shapes, thereby enhancing its adaptability to heterogeneous cropland structures and increasing the overall mapping precision.**

***Index Terms*—Boundary guidance, convolutional neural network (CNN)-transformer, cropland field parcels (CFPs), deep learning, high-resolution remote sensing imagery, spatial transferability.**

## I. Introduction

AGAINST the backdrop of an increasingly severe global food security situation, the development of smart agriculture and precision agriculture faces higher demands, posing new challenges for the refined management of cropland resources and the efficient acquisition of cropland information [1], [2], [3]. Cropland, as the core resource of agricultural production, is the foundation for ensuring agricultural development and food security, with its quantity and quality directly affecting the sustainability of agriculture [4], [5]. Cropland field parcels (CFPs) refer to spatially contiguous cropland units enclosed by distinct physical boundaries such as ridges, ditches, roads, or shelterbelts, which serve as the basic management unit for agricultural production. The boundary and spatial distribution information of CFPs is crucial for the implementation of precision agriculture [6], and the timely and accurate acquisition of this information holds significant importance for refined farmland management, efficient allocation of agricultural resources, and intelligent decision support.

Remote sensing technology, characterized by its wide coverage, strong timeliness, and rich information dimensions, has been widely applied in CFP mapping [7], [8], [9], [10]. In particular, the continuous launch of high-resolution remote sensing satellites has provided a solid data foundation for the fine delineation of parcel boundaries [11], [12], [13], [14], [15]. Traditional CFP mapping from remote sensing imagery mainly relies on manual visual interpretation, which is inefficient and costly, making it difficult to meet the

Received 17 September 2025; revised 17 November 2025 and 10 December 2025; accepted 2 January 2026. Date of publication 6 January 2026; date of current version 21 January 2026. This work was supported in part by the National Natural Science Foundation of China under Grant U23A2020, in part by Hubei Provincial Natural Science Foundation of China under Grant 2024AFA032, and in part by the Scientific Research Project of Hubei Natural Resources Department under Grant ZRZY2025KJ01. *(Corresponding author: Hao Wu.)*

Junyang Xie, Hao Wu, Lanfa Liu, and Anqi Lin are with the College of Urban and Environmental Sciences, Central China Normal University, Wuhan 430079, China (e-mail: xjy959@mails.ccnu.edu.cn; haowu@ccnu.edu.cn; lanfa@ccnu.edu.cn; linanqi@mails.ccnu.edu.cn).

Wenbin Wu and Qiangyi Yu are with the Institute of Agricultural Resources and Regional Planning, Chinese Academy of Agricultural Sciences, Beijing 100081, China (e-mail: wuwenbin@caas.cn; yuqiangyi@caas.cn).

Liang Hong and Lihua He are with Hubei Provincial Geographic National Conditions Monitoring Center, Wuhan 430071, China (e-mail: hongliang@hbsdlgqjczx.com; helihua@hbsdlgqjczx.com).

Jaturong Som-ard is with the Department of Geography, Faculty of Humanities and Social Sciences, Mahasarakham University, Maha Sarakham 44150, Thailand (e-mail: Jaturong.s@msu.ac.th).

Digital Object Identifier 10.1109/TGRS.2026.3651284

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:08:14 UTC from IEEE Xplore. Restrictions apply.

demands of large-scale and automated CFP mapping [16]. With the development of computer technology, methods such as edge detection [17], region segmentation [18], and traditional machine learning [19] have gradually been introduced into CFP mapping. These approaches have improved automation to some extent; however, due to their strong dependence on feature representation and their limited capability to effectively model the spatial heterogeneity of complex terrain and parcel boundaries, they still exhibit insufficient generalization ability and accuracy in diverse cropland scenarios, falling short of meeting the high requirements for mapping accuracy and boundary integrity in practical applications [3], [18], [20], [21].

In recent years, deep learning has been widely applied to the mapping of CFPs from remote sensing imagery, due to its advantages in automatic feature extraction, end-to-end learning, and nonlinear fitting [22], [23], [24], [25], [26]. According to the different model architectures, the existing deep-learning models for CFP mapping can be broadly categorized into three types: 1) convolutional neural network (CNN)-based models; 2) transformer-based models; and 3) CNN-transformer hybrid models [27]. Among the different models, CNN-based models are effective at local feature extraction, enabling accurate identification of CFP edges and fine texture details. As a result, many studies have focused on CNN architectures such as UNet and FCN [28], [29]. For example, Zhang et al. [30] improved the PSPNet model to extract CFPs in regions including Heilongjiang, Hebei, Zhejiang, and Guangdong in China. Subsequently, Xu et al. [31] applied an improved depthwise separable convolution UNet model successfully in Guangping, Hebei. Moreover, Xie et al. [32] introduced an edge feature enhancement module to the UNet model to improve the boundary accuracy. Although these models have achieved a high accuracy in delineating regularly shaped parcels, they often lose boundary details when applied to fragmented or irregular CFPs. Furthermore, constrained by the receptive field of the convolution operations, CNNs have limitations in modeling long-range spatial dependencies and capturing the global shape structure of CFPs [33]. This is particularly problematic in large-scale or fragmented agricultural landscapes, where incomplete boundaries and shape distortions frequently occur. To overcome these limitations, transformer-based models have been introduced for CFP mapping, capitalizing on their strong global modeling capabilities. Through self-attention mechanisms, transformers can effectively capture the long-distance dependencies and semantic consistency within images, thereby enhancing the spatial structure representation and showing superior adaptability to complex conditions such as blurred boundaries and irregular parcel shapes [27]. Notably, Swin transformer-based architectures have shown promising results in this domain. Sheng et al. [34] used a Swin transformer with a pyramidal structure as the backbone to obtain more precise boundary information of CFPs; Zhong et al. [35] improved and optimized the Swin transformer model to construct the MSMTransformer model, which can better extract the features exhibiting significant scale variations in farmland; and Xu et al. [36] combined the Swin transformer model with atrous convolution modules to maintain a robust performance even in densely distributed CFP regions. However, the transformer-based models generally lack the capability to express local details in high-resolution images during modeling, resulting in some difficulties in accurately locating CFP boundaries [37]. To fully leverage the strengths of both CNN and transformer models, related research has attempted to organically combine the local texture modeling ability of a CNN with the global context modeling capability of a transformer, thereby achieving synergistic extraction of CFP shapes and boundary information. For example, Zhang et al. [27] combined CNN and transformer models to develop MATNet, which obtained promising results in the Dali area of China. Wu et al. [33] integrated CNN and transformer models to fuse the global and local information, achieving a good performance even in fragmented CFP areas. In spite of their progress in improving the mapping accuracy and robustness, the existing hybrid models still face challenges such as boundary blurring, parcel adhesion, and suboptimal feature fusion. Therefore, designing an effective strategy to fully integrate the global and local information remains key to achieving high-precision mapping of CFPs.

Moreover, although high-resolution remote sensing imagery can provide rich details of surface features, the increased complexity of the background poses significant challenges. Cropland is often spectrally and texturally similar to other land covers such as forest and grassland, leading to frequent confusion [38]. Meanwhile, cropland boundaries are often blurred, especially in areas with complex terrain or intricate planting structures, which easily results in parcel adhesion, boundary misclassification, and severely affects the accuracy and spatial integrity of the mapping results [39]. Models relying solely on single semantic segmentation tasks often struggle to accurately capture the boundary details between parcels and, in particular, show a limited capability to express edge regions under complex scenarios, thus neglecting the transitional zones and boundary breaks between adjacent parcels [40], [41]. To address this issue, recent research has increasingly adopted multitask learning (MTL) approaches that jointly model the multiple auxiliary tasks related to the main task, aiming to enhance the overall representation ability and segmentation accuracy. For example, Xia et al. [42] distinguished CFP boundary regions into hard and soft edges and incorporated parcel extent constraints in a multitask collaborative framework, achieving improved CFP mapping results. In addition, models such as BsiNet [43], SEANet [44], and FPENet [32] learn the CFP extent, boundary, and distance-related tasks jointly, which enhances the boundary awareness while effectively alleviating issues such as parcel adhesion, edge gaps, and boundary displacement. However, some of the existing methods still fall short of fully exploiting the guiding value of boundary information in fine-grained segmentation, leading to local breaks and blurred contours in the final parcel boundaries. Therefore, it is imperative to design a deep network that integrates MTL with efficient feature fusion mechanisms, enabling the full exploitation of boundary information to improve the segmentation accuracy and achieve precise delineation of CFP boundaries while preserving their complete spatial structure.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:08:14 UTC from IEEE Xplore. Restrictions apply.

To address the aforementioned challenges, we propose a novel CNN-transformer hybrid network with boundary guidance (CTHBNet) for effective CFP mapping from high-resolution remote sensing imagery. CTHBNet is based on an encoder-decoder architecture. In the encoder, parallel dual CNN and transformer branches are employed to extract the local and global features of CFPs, respectively. This dual-branch structure enables the model to effectively capture both fine-grained spatial details (through the CNN) and long-range dependencies (through the transformer), thus offering a more comprehensive representation of CFPs. An effective CNN-transformer fusion module (CTF-Module) is also designed to integrate the comprehensive multiscale parcel information, thereby enhancing the mapping accuracy while preventing the loss of small and elongated CFPs. In the decoder, a hierarchical information fusion decoder decoding module (HIF-Module) is introduced, which progressively restores the parcel boundaries through DUpsampling and multiscale feature fusion. In addition, a boundary-guided feature enhancement module (BGFE-Module) is designed, which optimizes the boundary representation by combining the Sobel gradient operator with a global attention mechanism, thereby effectively mitigating issues such as parcel adhesion, omission, and misclassification. Moreover, an MTL strategy integrating the parcel extent, distance, and boundary tasks is adopted to further improve the precision and robustness of CFP mapping. The performance and accuracy of the proposed CTHBNet model were evaluated on datasets from four regions in China and five regions worldwide, along with three existing benchmark datasets, demonstrating its effectiveness in CFP mapping.

The main contributions of this article are summarized as follows.

1) We propose CTHBNet, a novel deep-learning model that integrates CNN and Transformer with boundary guidance, enabling more accurate mapping of CFPs from high-resolution remote sensing imagery.
2) We design a CTF-Module for parallel fusion of CNN and Transformer features, facilitating complementary learning of global and local CFP representations. Meanwhile, a BGFE-Module is proposed to enhance boundary feature recognition, improving the integrity of CFP mapping results.
3) We validate the superior CFP mapping capability of CTHBNet across four distinct agricultural landscape regions in China. Comparative experiments demonstrate its higher accuracy and robustness, while ablation studies confirm the contribution of each proposed module to overall mapping accuracy and boundary delineation.
4) Experiments on benchmark datasets further demonstrate the strong performance of CTHBNet in CFP mapping, and extensive validation across five global regions highlights its superior transferability and generalization.

## II. Materials

### A. Study Area

We selected four representative regions in China to evaluate the performance of the proposed CTHBNet model in CFP mapping (Fig. 1). Specifically, Region A is located in the city of Dangyang, Hubei province. Situated in the Jianghan Plain of central China, Dangyang has a subtropical monsoon climate with abundant rainfall and heat. The city has approximately 807.6 $km^2$ of arable land, accounting for 37.57% of its total land area. The terrain is complex, featuring a dense network of rivers and scattered lakes. The agricultural landscape is highly fragmented, with the cropland parcels varying significantly in size and shape, posing considerable challenges for accurate boundary mapping.

TABLE I
GF-2 Data of the Four Study Regions

| Regions | Dates (Scenes) |
|---|---|
| Region A (Hubei) | 2022/2/23 (2), 2022/3/5 (2), 2023/1/29 (1), 2023/6/26 (8) |
| Region B (Shandong) | 2022/04/23 (1) |
| Region C (Jilin) | 2021/10/11 (1) |
| Region D (Chongqing) | 2021/05/09 (1) |

Region B is the city of Dezhou in Shandong, located in the Huang-Huai-Hai Plain. CFPs in this area are generally large and regularly shaped, with clear spectral distinction from the surrounding features. Region C is the city of Yushu in Jilin, which is part of the Northeast China Plain. The area is flat and open, which is favorable for large-scale mechanized farming, with relatively large and well-defined CFPs. Region D is Nanchuan in Chongqing municipality, located in a mountainous area of southwestern China. This region has highly undulating terrain, and the farmland is dominated by terraced fields that are elongated, irregular, and spatially fragmented.

A comparative analysis of the CFPs across the four regions shows that the average parcel areas in Regions A–D are 0.21, 4.37, 0.70, and 0.02 ha, respectively, while the average shape regularity scores are 0.76, 0.88, 0.86, and 0.41. These results indicate the significant differences among the four regions in terms of parcel scale, shape complexity, and geomorphological background. Such diversity effectively reflects the varying cropland distribution patterns under different agricultural landscapes, thereby supporting a comprehensive evaluation of the proposed model's adaptability and robustness.

### B. Remote Sensing Data Collection and Pre-Processing

We obtained Gaofen-2 (GF-2) remote sensing imagery from the Land Observation Satellite Data Service Platform of the China Centre for Resources Satellite Data and Application. Specifically, 13 GF-2 images were mosaicked and cropped to generate the dataset for Region A, while one single image was used for each of Regions B–D. The cloud cover in the original data for all images does not exceed 5%. Detailed information is provided in Table I. The GF-2 satellite, which was successfully launched on August 19, 2014, is China's first independently developed civilian optical remote sensing satellite with a spatial resolution better than 1 m. GF-2 provides one panchromatic band (450–900 nm) and four multispectral bands: blue (450–520 nm), green (520–590 nm), red (630–690 nm), and near-infrared (770–890 nm).

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:08:14 UTC from IEEE Xplore. Restrictions apply.

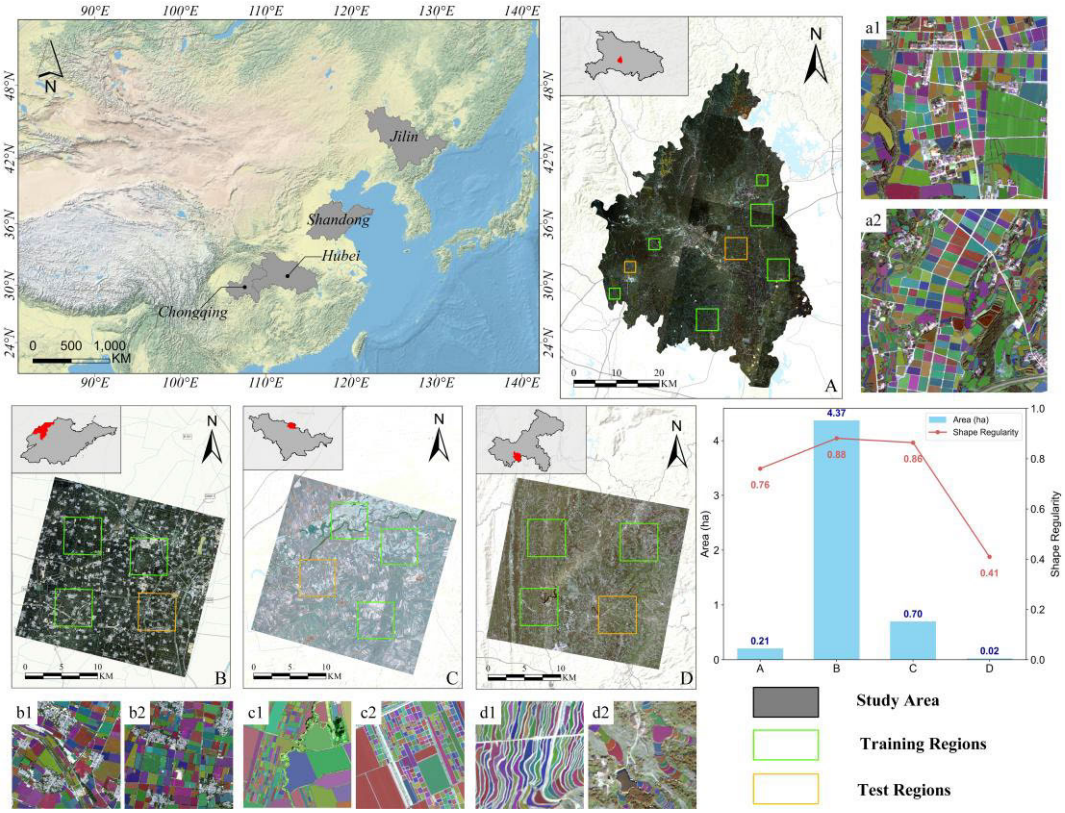



Fig. 1. Spatial locations of the selected study areas. (A)–(D) Four study areas located in Hubei, Chongqing municipality, Shandong, and Jilin, respectively. The green and yellow boxes indicate the spatial extents of the training and testing regions. The eight panels (a1), (a2), (b1), (b2), (c1), (c2), (d1), and (d2) are representative examples showing the various agricultural field shapes and sizes over the four study areas.

During the data preprocessing, the panchromatic and multispectral bands of the GF-2 imagery were first subjected to radiometric calibration, atmospheric correction, and geometric correction using ENVI software. Subsequently, the NNDiffuse pan-sharpening method was applied to fuse the panchromatic and multispectral bands, producing GF-2 multispectral images with a spatial resolution of 1 m. Since Region A required multiple scenes for full coverage, image mosaicking was performed to obtain a complete GF-2 multispectral dataset covering the entire study area.

### C. CFP Annotation

The input data for CTHBNet included the preprocessed remote sensing imagery and the corresponding sample label data. The remote sensing data were derived from GF-2 multispectral images (red, green, and blue bands) with the bit depth rescaled to 8-bit, while the labels contained information on parcel extent, boundaries, and distance. During the training and validation phase, Region A included three sample areas of $5120 \times 5120$ pixels and three sample areas of $2560 \times 2560$ pixels, while Regions B–D each included three sample areas of $5120 \times 5120$ pixels. In the testing phase, Region A contained one $5120 \times 5120$ pixels test area and one $2560 \times 2560$ pixels test area, and Regions B–D each included one $5120 \times 5120$ pixels test area. This diverse and large-scale dataset enabled a comprehensive and robust evaluation of the model's performance.

The sample annotation in this study was collaboratively performed by five experienced interpreters. Cross-validation was employed to ensure the accuracy of the labeled data, resulting in vector data representing the extent of the CFPs, which were then converted into binary raster images. Subsequently, the Canny edge detection and distance transform algorithms from the OpenCV library were used to extract the boundary and distance information of the farmland parcels. Specifically, the Canny edge detector (cv2.Canny) was configured with a low threshold of 30 and a high threshold of 255 to extract clear boundaries. The distance transform (cv2.distanceTransform) was performed using the Euclidean distance (DIST_L2) with a mask size of 3 to compute the distance from each pixel to the nearest edge. The final input data for the model are illustrated in Fig. 2.

### D. Generalization Validation Area and Data

To further evaluate the transferability and generalization capability of the proposed model, experiments were conducted in one representative region from each of the following continents: Europe, Africa, Oceania, North America, and South America. The geographical locations of these regions are shown in Fig. 3. Specifically, T1 is located in Flevoland, the Netherlands (Europe); T2 is in Mpumalanga, South Africa (Africa); T3 is in Victoria, Australia (Oceania); T4 is in California, United States (North America); and T5 is in Maule, Chile (South America).

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:08:14 UTC from IEEE Xplore. Restrictions apply.

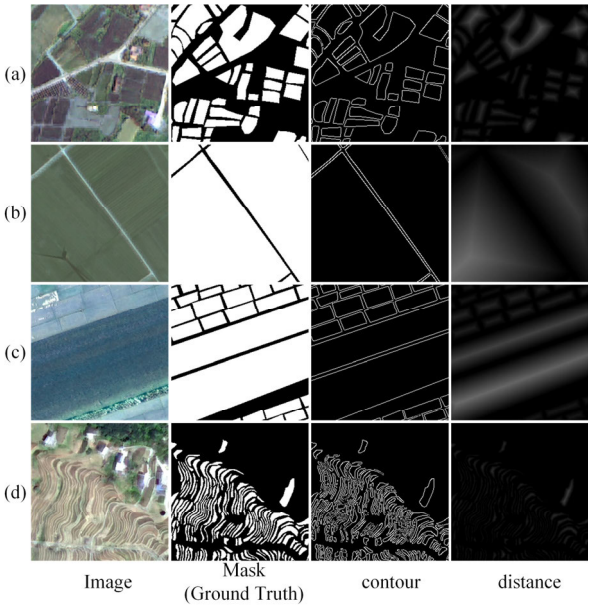



Fig. 2. Sample datasets for CFP extraction. (a)–(d) show detailed sample datasets for Dangyang (Hubei Province), Dezhou (Shandong Province), Yushu (Jilin Province), and Nanchuan (Chongqing Municipality), respectively.

Region T1 is located in central Netherlands, which is an area formed through land reclamation. The region features low-lying terrain and large, contiguous farmland plots with regular geometric shapes. Agriculture in this area is highly mechanized, reflecting intensive and modern farming practices. Region T2 is situated in northeastern South Africa, characterized by a subtropical climate and complex topography, sloping from east to west. The region is dominated by smallholder farms, and the farmland plots are irregular due to the diverse terrain. T3 lies in southeastern coastal Australia, where agriculture and animal husbandry are well developed. The CFPs in this region are generally large and rectangular; however, influenced by the temperate oceanic climate and saline-alkaline soils, some of the field boundaries appear blurred. Region T4, located in California, United States, is dominated by large-scale commercial farms with diverse agricultural types and expansive field sizes. Region T5 is in central Chile, primarily consisting of flat valley terrain. The region supports medium- to large-scale farming operations suitable for intensive cultivation, with a wide variety of crops.

In addition, the statistical analysis of the area and shape regularity index of cropland parcels across the five regions (T1–T5) provides further insights into the performance differences. As shown in Fig. 3, the average cropland parcel area for regions T1–T5 is as follows: 3.96 ha (T1), 8.71 ha (T2), 8.87 ha (T3), 10.02 ha (T4), and 2.55 ha (T5). This indicates a clear trend where T1 and T5 have smaller average parcel sizes, while T4 has the largest. Smaller parcels in T1 and T5 likely contribute to fragmentation and irregular shapes, making it more challenging for the model to accurately delineate boundaries. In contrast, the larger parcel sizes in T4 suggest fewer segmentation difficulties due to the presence of more continuous and regularly shaped fields. Furthermore, the average shape regularity index for each region is as follows: 0.87 (T1), 0.70 (T2), 0.82 (T3), 0.88 (T4), and 0.79 (T5). The shape regularity index reflects the geometric regularity of the parcels, with higher values indicating more regular and simpler shapes. The lower shape regularity in T2 (0.70), along with the smaller parcel size, suggests that this region has more irregular and fragmented parcels, which can hinder the model's ability to extract boundaries accurately. On the other hand, T4, with both larger and more regularly shaped parcels (shape regularity of 0.88), demonstrates better segmentation performance. In saummary, these five international regions span diverse landscapes, climatic zones, and agricultural systems, making them representative and suitable for thoroughly evaluating the transferability and generalization capacity of the proposed model.

TABLE II
IMAGE INFORMATION FOR THE FIVE MIGRATION EXPERIMENT REGIONS

| Region | Resolution | Date | Coverage (pixels) |
|---|---|---|---|
| T1 (Europe) | 3 m | 2020/05/09 | 36079 × 33126 |
| T2 (Africa) | 1 m | 2019/11/24 | 10240 × 10240 |
| T3 (Oceania) | 1 m | 2018/12/01 | 10240 × 10240 |
| T4 (North America) | 1 m | 2019/11/24 | 10240 × 10240 |
| T5 (South America) | 1 m | 2020/10/16 | 10240 × 10240 |

Table II presents detailed information on the datasets for each region, all obtained from Google Earth (Google Inc.). The datasets consist of three spectral bands: Red, Green, and Blue (RGB), and all of them are cloud-free. For the sample annotation, the CFPs in Region T1 were sourced from the official Dutch geospatial data platform PDOK. For the remaining regions, the annotation was manually conducted by five experienced interpreters based on the visual interpretation of satellite imagery, ensuring high annotation accuracy and reliability. In each region, three image patches of $5120 \times 5120$ pixels were selected as training areas, and one patch of the same size was used as the testing area.

## III. METHODS

### A. Proposed CTHBNet Architecture

The architecture of the proposed CTHBNet model is shown in Fig. 4. CTHBNet consists of three main modules: CTF-Module, HIF-Module, and BGFE-Module. The CTF-Module integrates the strengths of a CNN model in capturing local spatial details and the ability of a transformer model to model long-range dependencies, enabling the effective extraction of cropland features from both local and global perspectives. The HIF-Module combines high-level semantic representations from the deeper layers with detailed spatial features from the shallower layers, progressively restoring the spatial resolution while preserving both the semantic and positional information to support precise segmentation. The BGFE-Module extracts directional texture cues from the original imagery in both horizontal and vertical directions and performs multiplicative fusion with cropland features. This enhances the boundary contrast between cropland and adjacent land covers and strengthens the cropland-relevant features from a global

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:08:14 UTC from IEEE Xplore. Restrictions apply.

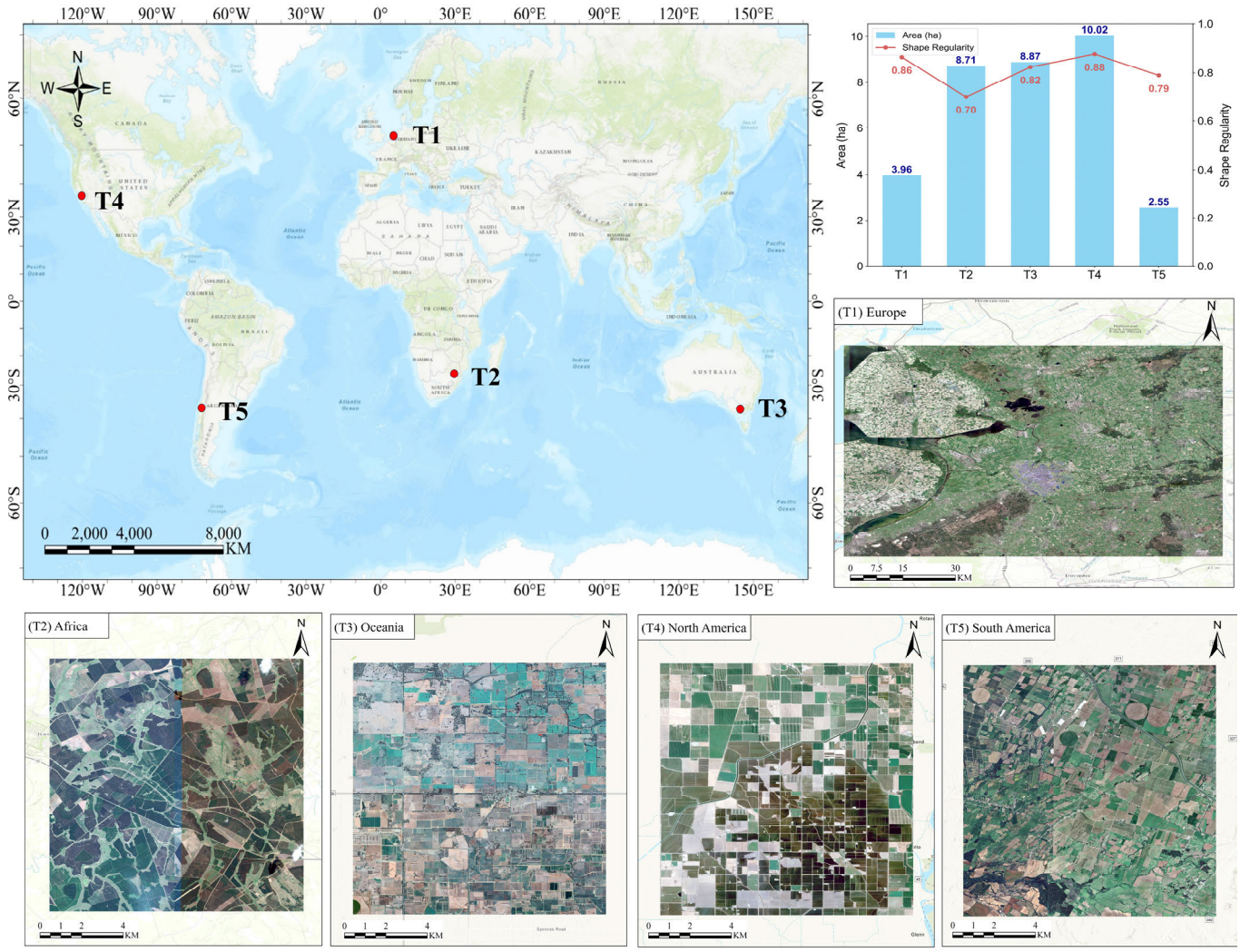



Fig. 3. Spatial location of the selected global migration experiment regions. T1–T5 refer to the five regions located in the Netherlands in Europe, South Africa in Africa, Australia in Oceania, the United States in North America, and Chile in South America, respectively.

context. Additionally, CTHBNet adopts a MTL strategy, which jointly models region, boundary, and distance constraints. This strategy helps optimize the segmentation performance, ensuring more accurate and complete CFP mapping. By addressing multiple aspects of CFP extraction in parallel, CTHBNet improves both the precision and robustness of the model. The detailed structures of each module are presented in Sections III-A1–III-A4.

*1) CTF-Module:* In CFP mapping tasks based on semantic segmentation, the boundaries between cropland and the surrounding land covers are often ambiguous, particularly in areas with complex terrain or heterogeneous agricultural patterns. This ambiguity can lead to incomplete segmentation results, where adjacent parcels are not clearly separated. Moreover, factors such as spectral similarity between neighboring parcels or sub-pixel-level spacing between field edges often result in adhesion or merging of parcels in the output. In this context, CNN models are effective at capturing local spatial features, such as textures and edges, which are essential for accurate boundary delineation. On the other hand, transformer models leverage self-attention mechanisms to model the long-range dependencies, making them well suited for maintaining the overall structural integrity of CFPs. By combining the respective strengths of both architectures, the fusion model enhances the ability to detect parcel boundaries while also improving the adaptability to varying parcel shapes and sizes, thereby reducing adhesion and misclassification. To this end, we designed the CTF-Module, which integrates the strengths of a CNN model in capturing local spatial details with the transformer's ability to model long-range dependencies. By combining these two approaches, CTHBNet effectively extracts cropland features from both local and global perspectives. This hybrid fusion allows for a richer representation of CFPs, overcoming the limitations of models that rely solely on CNNs or transformers.

In the CNN branch, a bottleneck block is employed [Fig. 5(a)] [45] to effectively extract local features such as textures and edges. The deep residual connections within the module help mitigate the vanishing gradient problem while preserving fine-grained spatial details. A four-level multi-scale encoding structure is adopted, enabling the network to learn cropland boundaries and texture information at dif-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:08:14 UTC from IEEE Xplore. Restrictions apply.

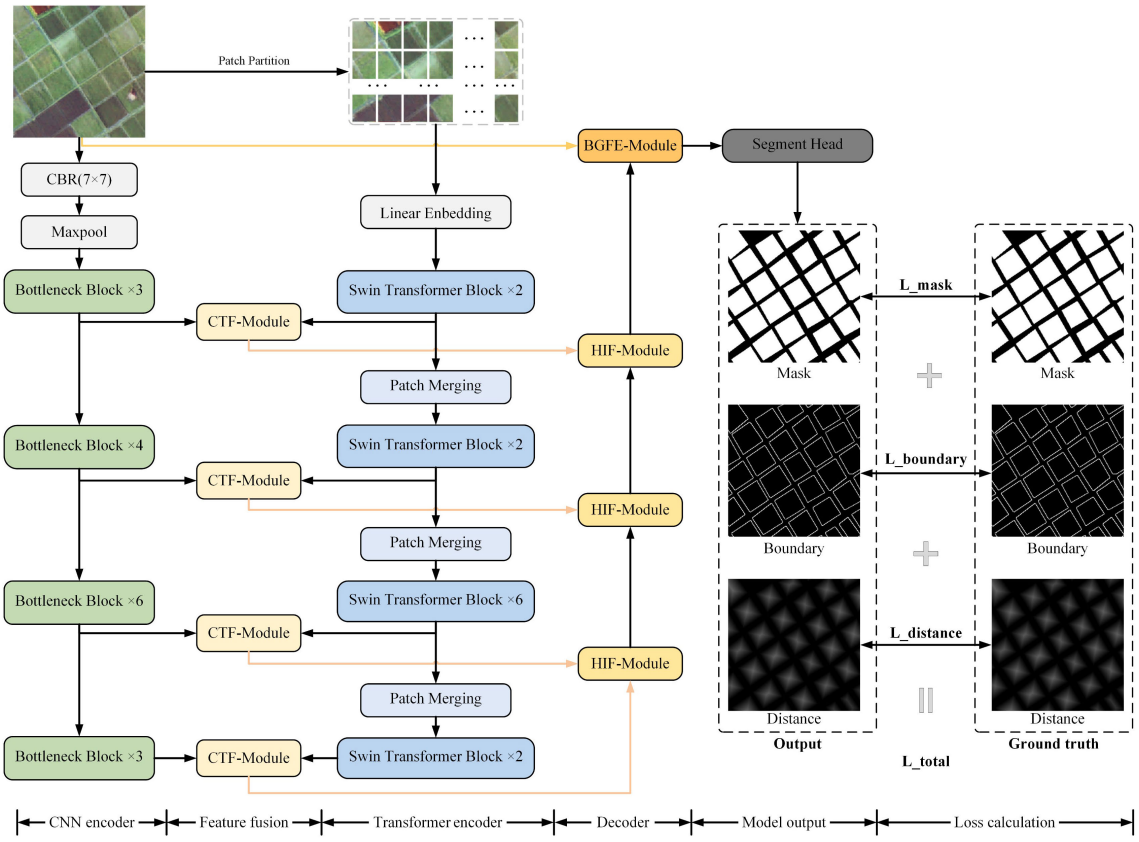



Fig. 4. Main architecture of CTHBNet. CTF-Module, HIF-Module, and BGFE-Module refer to the CNN-transformer fusion module, hierarchical information fusion decoder module, and boundary-guided feature enhancement module, respectively.

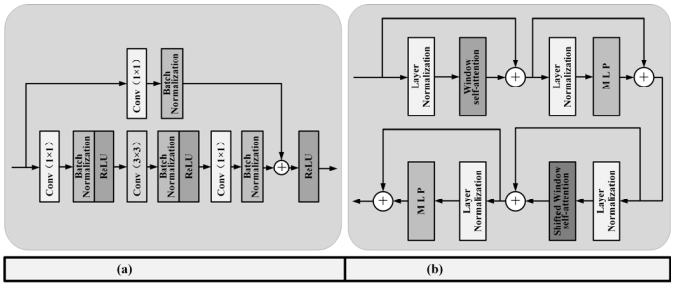



Fig. 5. Architecture of (a) bottleneck block and (b) Swin transformer block.

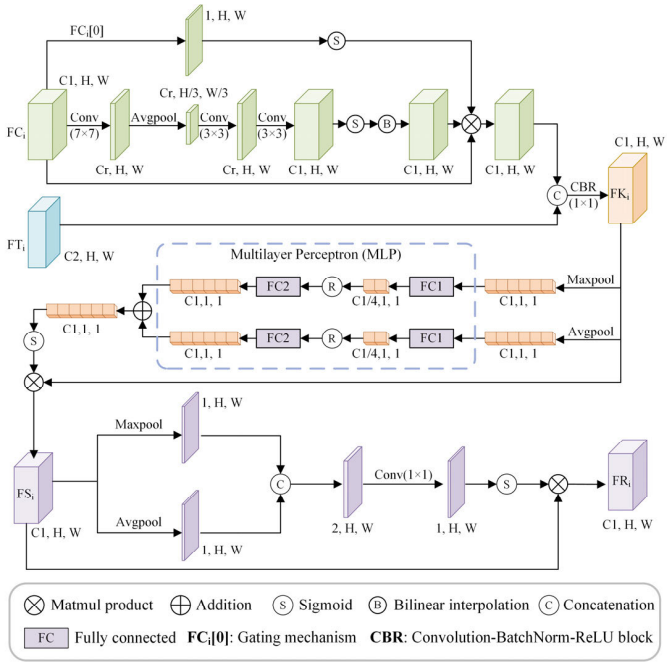



Fig. 6. Architecture of the HIF-Module.

ferent scales, thereby facilitating fine-grained segmentation of irregularly shaped parcels. Meanwhile, in the transformer branch, a Swin transformer block [Fig. 5(b)] [46] is used. By leveraging a self-attention mechanism, it captures the long-range dependencies and demonstrates a strong performance in handling large-scale cropland parcels and complex terrain conditions. Furthermore, the hierarchical window-based attention mechanism allows the model to simultaneously capture both local and global contextual information across multiple scales, which helps preserve the structural integrity of the CFP shapes.

The structure of the CTF-Module is illustrated in Fig. 6. In this module, a local importance attention mechanism is applied to the CNN branch to enhance the model's sensitivity to critical local features while suppressing interference from irrelevant background regions. The features extracted from the CNN and transformer branches are then stacked and fused, followed by a Convolution-BatchNorm-ReLU (CBR) module that compresses the feature dimensions and improves the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:08:14 UTC from IEEE Xplore. Restrictions apply.

stability of the feature distribution. Finally, feature enhancement is performed across both the channel and spatial dimensions, further optimizing the fused representation and guiding the model to focus on the most informative regions within the imagery.

The detailed computational process is described as follows:

$$\mathrm{FR}_i = \mathrm{FS}_i \times \sigma\left(\mathrm{Conv}_{1\times1}\left(\mathrm{Concat}\left(\mathrm{Maxpool}\left(\mathrm{FS}_i\right), \mathrm{Avgpool}\left(\mathrm{FS}_i\right)\right)\right)\right) \tag{1}$$

$$\mathrm{FS}_i = \mathrm{FK}_i \times \sigma\left(\mathrm{Mlp}\left(\mathrm{Maxpool}\left(\mathrm{FK}_i\right)\right) + \mathrm{Mlp}\left(\mathrm{Avgpool}\left(\mathrm{FK}_i\right)\right)\right) \tag{2}$$

$$\mathrm{FK}_i = \mathrm{CBR}_{1\times1}\left(\mathrm{Concat}\left(\mathrm{FT}_i, \mathrm{FCZ}_i\right)\right) \tag{3}$$

$$\mathrm{FCZ}_i = \mathrm{FC}_i \times \sigma\left(\mathrm{FC}_i\left[0\right]\right) \times \mathrm{interpolation}\left(\sigma\left(\mathrm{Conv}_{3\times3}\left(\mathrm{Conv}_{3\times3}\left(\mathrm{Avgpool}\left(\mathrm{Conv}_{7\times7}\left(\mathrm{FC}_i\right)\right)\right)\right)\right)\right). \tag{4}$$

Specifically, the CNN feature map $\mathrm{FC}_i(i = 1, 2, 3, 4) \in \mathbb{R}^{C1\times H\times W}$ is first processed through a gating mechanism, followed by a nonlinear transformation using the Sigmoid activation function to generate an attention map $\mathrm{FC1}_i(i = 1, 2, 3, 4) \in \mathbb{R}^{1\times H\times W}$. Simultaneously, $\mathrm{FC}_i$ is passed through a convolutional layer with a $7 \times 7$ kernel, followed by average pooling. The output is then convolved with two successive $3 \times 3$ convolutional layers and activated by a Sigmoid function, after which bilinear interpolation is applied to obtain another attention feature map $\mathrm{FC2}_i(i = 1, 2, 3, 4) \in \mathbb{R}^{C1\times H\times W}$. $FC_i$, $FC1_i$, and $FC2_i$ are then multiplied element-wise, and the result is concatenated with the transformer feature map $\mathrm{FT}_i(i = 1, 2, 3, 4) \in \mathbb{R}^{C2\times H\times W}$. The fused features are subsequently passed through a CBR module with a $1 \times 1$ convolutional kernel to generate the intermediate feature representation $\mathrm{FK}_i(i = 1, 2, 3, 4) \in \mathbb{R}^{C1\times H\times W}$. Next, $\mathrm{FK}_i$ undergoes both channel-wise max pooling and average pooling, and the pooled features are separately passed through a multilayer perceptron (MLP). The resulting features are added together, activated by a Sigmoid function, and then multiplied by the original $\mathrm{FK}_i$ to produce the enhanced feature map $\mathrm{FS}_i(i = 1, 2, 3, 4) \in \mathbb{R}^{C1\times H\times W}$. Finally, spatial-wise max pooling and average pooling are applied to $\mathrm{FS}_i$, and their outputs are concatenated along the channel dimension. The concatenated feature is then passed through a $1\times1$ convolution followed by a Sigmoid activation function. The resulting attention map is multiplied by $\mathrm{FS}_i$, yielding the final fused feature representation $\mathrm{FR}_i(i = 1, 2, 3, 4) \in \mathbb{R}^{C1\times H\times W}$, which effectively integrates both the CNN and transformer features.

*2) HIF-Module:* In semantic segmentation models with an encoder-decoder architecture, the encoder is responsible for extracting multiscale cropland features, while the decoder gradually recovers the spatial details to reconstruct the cropland parcel boundaries. Conventional decoding operations, such as transposed convolutions and interpolation, face several challenges. On the one hand, transposed convolutions perform upsampling with fixed strides and kernel sizes, which may introduce grid-like artifacts, which are commonly referred to as the "checkerboard effect." These artifacts disrupt the spatial continuity between cropland parcels and result in unrealistic boundaries that severely compromise the completeness and accuracy of CFP mapping. On the other hand, standard interpolation-based upsampling often leads to blurred boundaries and the loss of fine-grained details, particularly affecting small or elongated parcels. This can cause adhesion between neighboring parcels or even the complete omission of narrow cropland regions, posing significant challenges for precise cropland mapping.

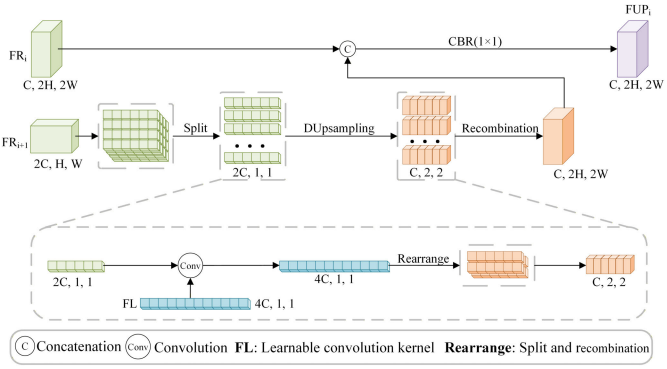



Fig. 7. Architecture of the CTA-Module.

To address these issues, we propose the HIF-Module, as illustrated in Fig. 7. This module progressively restores the spatial resolution by combining high-level semantic representations from deeper layers with detailed spatial features from shallower layers. This hierarchical fusion of features ensures that both semantic and positional information are preserved, enabling precise segmentation even in areas with complex land structures. This module's ability to integrate features at multiple scales and resolutions sets CTHBNet apart from simpler models that do not leverage multilevel feature fusion. It employs DUpsampling to restore the spatial resolution of the high-level semantic features, aligning them with the resolution of the low-level features. The restored high-level features are then concatenated with the corresponding low-level features to integrate global semantic context with rich local detail. This hierarchical fusion strategy effectively reduces information loss and enhances the quality of the decoded representation. Finally, the CBR module with a $1 \times 1$ convolutional kernel is applied to compress the number of feature channels, which prevents computational redundancy while maintaining efficient information transmission. The detailed computational process is as follows:

$$\mathrm{FUP}_i = \mathrm{CBR}_{1\times1}\left(\mathrm{Concat}\left(\mathrm{DU}\left(\mathrm{FR}_{i+1}\right), \mathrm{FR}_i\right)\right) \tag{5}$$

$$\mathrm{DU}\left(\mathrm{FR}_{i+1}\right) = \mathrm{Rrarrange}\left(\mathrm{Split}\left(\mathrm{FR}_{i+1}\right) \times \mathrm{FL}\right). \tag{6}$$

Specifically, the high-level feature map $\mathrm{FR}_{i+1}(i = 1, 2, 3) \in \mathbb{R}^{2C\times H\times W}$ is first split along the spatial dimensions to generate $H \times W$ features $\mathrm{FR1}_{i+1}(i = 1, 2, 3) \in \mathbb{R}^{2C\times1\times1}$, each corresponding to a spatial location. A learnable convolutional kernel $FL \in \mathbb{R}^{4C\times1\times1}$ is then applied to each of them through convolution operations, producing a set of $H \times W$ transformed features $\mathrm{FR2}_{i+1}(i = 1, 2, 3) \in \mathbb{R}^{4C\times1\times1}$. These features are subsequently divided along the channel dimension and reorganized along the spatial dimensions to form $H \times W$ features $\mathrm{FR3}_{i+1}(i = 1, 2, 3) \in \mathbb{R}^{C\times2\times2}$. After rearrangement and aggregation, the fused feature map $\mathrm{FR4}_{i+1}(i = 1, 2, 3) \in \mathbb{R}^{C\times2H\times2W}$ is obtained. Finally, $\mathrm{FR4}_{i+1}$ and $\mathrm{FR}_i$ are concatenated along the channel dimension. The concatenated features are passed through the CBR module with a $1 \times 1$ convolutional kernel,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:08:14 UTC from IEEE Xplore. Restrictions apply.

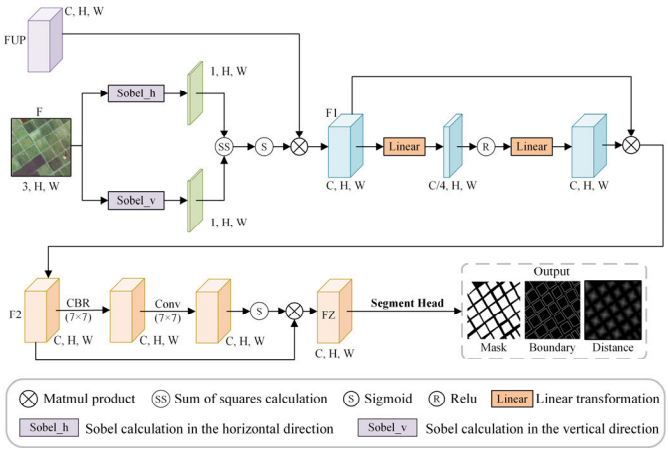



Fig. 8. Architecture of the BGFE-Module.

yielding the output $\mathrm{FUP}_i (i = 1, 2, 3) \in \mathbb{R}^{C\times 2H\times 2W}$ of the HIF-Module.

In the HIF-Module, DUpsampling [47] is employed to upsample the high-level features. Learnable convolution is first used to directly double the number of channels in the original high-dimensional features. These features are then split along the channel dimension and reorganized in the spatial domain to restore the spatial resolution, aligning it with that of the corresponding low-level features. This upsampling strategy effectively avoids the checkerboard artifacts and detail loss typically caused by transposed convolution or interpolation, thereby preserving the accuracy of the cropland boundary regions. Moreover, by fusing the restored high-level features with the low-level features, the model incorporates detailed local boundary information. This fusion significantly enhances the clarity and completeness of the field boundaries, resulting in more accurate and robust CFP mapping.

*3) BGFE-Module:* Because of the significant variations in agricultural landscapes across different regions, CFPs often exhibit considerable diversity in shape and size. While the DUpsampling-based upsampling method and effective feature fusion in the decoder improve the clarity and completeness of the parcel boundaries, challenges remain when dealing with small and narrow cropland parcels. These challenges include the potential loss of fine-grained features and insufficient spatial structure representation.

To address these limitations and further enhance boundary awareness, we propose the BGFE-Module, which leverages boundary information to guide the refinement of the decoded features, aiming to improve both the boundary completeness and classification accuracy. The structure of the module is shown in Fig. 8. The module first extracts edge features from the input image using the Sobel operator, which enhances the boundary cues through gradient-based edge detection. These edge features are then fused with the high-level features from the decoder to reinforce the boundary representations. However, boundary features are often sensitive to noise, and direct element-wise multiplication with the decoded features may lead to over-enhancement or discontinuities along parcel edges. To mitigate these issues, the fused features are further refined at a global level to optimize the boundary consistency and preserve fine details. This process ensures improved mapping accuracy for CFPs, especially in complex agricultural landscapes. A core innovation of CTHBNet, the BGFE-Module extracts directional texture cues from the original imagery in both horizontal and vertical directions. These texture features are then fused multiplicatively with the extracted cropland features, enhancing the boundary contrast between cropland and adjacent land covers. The integration of global context through a global attention mechanism strengthens the cropland-relevant features, making this module particularly effective for accurate boundary delineation in complex agricultural landscapes. The detailed computational process is described as follows:

$$F1 = \sigma\left(\sqrt{(\mathrm{Sobel_}h(F))^2 + (\mathrm{Sobel_}v(F))^2}\right) \times \mathrm{FUP} \quad (7)$$

$$F2 = \mathrm{Linear}\left(\mathrm{Relu}\left(\mathrm{Linear}\left(F1\right)\right)\right) \times F1 \quad (8)$$

$$FZ = \sigma\left(\mathrm{Conv}_{7\times 7}\left(\mathrm{Conv}_{7\times 7}\left(F2\right)\right)\right) \times F2. \quad (9)$$

Specifically, the original input image $F \in \mathbb{R}^{3\times H\times W}$ is first processed with the Sobel operator in both horizontal and vertical directions, and the resulting gradient maps are fused using the sum of squared responses to generate a boundary map $F' \in \mathbb{R}^{3\times H\times W}$ that highlights the cropland parcel edges. After applying Sigmoid activation to $F' \in \mathbb{R}^{3\times H\times W}$, it is element-wise multiplied by $\mathrm{FUP} \in \mathbb{R}^{C\times H\times W}$ to produce $F1 \in \mathbb{R}^{C\times H\times W}$, thereby enhancing the features near cropland parcel boundaries and guiding the model to focus more on these boundary regions. Next, $F1$ undergoes a linear transformation followed by a rectified linear unit (ReLU) activation and another linear transformation, and the resulting feature map is then element-wise multiplied with $F1$ to produce the refined feature map $F2 \in \mathbb{R}^{C\times H\times W}$. Finally, $F2$ is processed by two consecutive $7 \times 7$ convolutional layers followed by Sigmoid activation, and then element-wise multiplied with itself to produce the module's output feature $FZ \in \mathbb{R}^{C\times H\times W}$. This enables the model to dynamically adjust the influence of the boundary information globally, enhancing the critical features while suppressing the noise in nonboundary regions.

*4) MTL:* As a form of inductive transfer, MTL is a widely adopted strategy in deep learning that simultaneously addresses multiple related tasks by sharing the learned knowledge to improve the performance across each task. By jointly training on multiple tasks, the model leverages both the commonalities and differences among the tasks, learning their interrelations to enhance the generalization ability and better adapt to complex and diverse scenarios.

In the proposed model, we employ an MTL strategy to improve the accuracy and boundary clarity of the CFP mapping. Specifically, the model's segmentation head concurrently predicts the parcel's extent, boundary, and distance maps. These predictions are supervised using the original labels, which include corresponding information on extent, boundary, and distance, enabling the model to better perceive the structural and morphological characteristics of the parcels. The extent prediction focuses on delineating the main body of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:08:14 UTC from IEEE Xplore. Restrictions apply.

the cropland parcels to ensure accurate recognition of parcels at various scales. The boundary prediction leverages gradient information to learn edge features, thereby enhancing the model's sensitivity to parcel boundaries and reducing issues such as parcel adhesion and boundary blurring. The distance prediction estimates the distance from each pixel to the nearest parcel boundary, providing additional geometric constraints that strengthen the model's adaptability to parcels of diverse shapes and sizes.

*5) Loss Function:* For the loss calculation, supervision is applied separately to the extent, boundary, and distance predictions. The individual losses are computed and then summed to form the total loss function $L_\text{total}$ of the model, as expressed in the following equation:

$$L_\text{total} = L_\text{mask} + L_\text{boundary} + L_\text{distance} \tag{10}$$

where $L_\text{mask}$ denotes the extent loss calculated by binary cross-entropy loss [see (11)], $L_\text{boundary}$ represents the boundary loss computed using dice loss [see (12)], and $L_\text{distance}$ corresponds to the distance loss measured by Jaccard distance loss [see (13)]. Considering the overall objective, all the tasks are assigned equal importance; thus, the total loss function is the sum of these three losses

$$L_{\text{BCE}} = -\left(y_1 \times \log\left(\hat{y}_1\right) + \left(1 - y_1\right) \times \log\left(1 - \hat{y}_1\right)\right) \tag{11}$$

$$L_{\text{Dice}} = 1 - \frac{2|\text{y}_2 \cap \hat{\text{y}}_2|}{|\text{y}_2| + |\hat{\text{y}}_2|} \tag{12}$$

$$L_{\text{Jaccard}} = \frac{|\text{y}_3 \cap \hat{\text{y}}_3|}{|\text{y}_3| + |\hat{\text{y}}_3| - |\text{y}_3 \cap \hat{\text{y}}_3|} \tag{13}$$

where $\text{y}_\text{i}$ denotes the ground-truth value, $\hat{\text{y}}_\text{i}$ represents the predicted value from the model, and $i = 1, 2, 3$ correspond to the cropland parcel extent, boundary, and distance, respectively.

### B. Experimental Settings

*1) Collection of the Training Samples:* In this study, the input data for the model were set to a size of $256 \times 256$ pixels. To ensure sufficient data for training and testing, and to reduce the risk of overfitting during model training [48], we employed data augmentation techniques such as image rotation, flipping, and the addition of random noise to expand the sample set. Furthermore, the training dataset was further split into training and validation sets at a ratio of 8:2. The final dataset composition is summarized in Table III.

*2) Parameter Settings:* The experiments were conducted on an Ubuntu 20.04 system using Python 3.8 and the PyTorch 1.10.0 deep-learning framework. Model training was performed on a single NVIDIA GTX 3060 GPU with 12 GB of memory. The initial learning rate was set to 0.0001, with a total of 100 training epochs and a batch size of 8. A MultiStepLR learning rate decay strategy was employed, where the learning rate was decreased by a factor of 10 at the 50th and 85th epochs. The Adam optimizer [49] was used to optimize the network parameters.

### C. Accuracy Assessment

Accuracy evaluation is a key indicator for assessing the generalization ability and performance of the model. In this study, we evaluated the precision of the CFP mapping results using several metrics derived from the confusion matrix: overall accuracy (OA), precision, recall, F1-score, and mean intersection over union (mIoU).

TABLE III
DATA USED IN THE EXPERIMENT

| Dataset | Region | Training data (Original/Expanded) | Test data (Original/Expanded) |
|---|---|---|---|
| GF-2 dataset | Hubei | 1500 / 4500 | 500 / 1500 |
| | Chongqing | 1200 / 4500 | 400 / 1500 |
| | Shandong | 1200 / 4500 | 400 / 1500 |
| | Jilin | 1200 / 4500 | 400 / 1500 |
| Validation dataset | Europe | 1200 / 3000 | 400 / 1000 |
| | Africa | 1200 / 3000 | 400 / 1000 |
| | Oceania | 1200 / 3000 | 400 / 1000 |
| | North America | 1200 / 3000 | 400 / 1000 |
| | South America | 1200 / 3000 | 400 / 1000 |

The OA represents the proportion of correctly predicted pixels of both cropland and noncropland classes over the total number of pixels. The precision measures the proportion of correctly predicted cropland pixels among all the pixels predicted as cropland. The recall indicates the proportion of correctly predicted cropland pixels relative to the total number of true cropland pixels in the ground truth. The F1-score is the harmonic mean of the precision and recall, providing a balanced measure of the model's accuracy and completeness. The mIoU calculates the ratio of the intersection to the union between the predicted and ground-truth cropland pixels. The specific formulas for each metric are as follows:

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{14}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{15}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{16}$$

$$F1 - \text{score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{17}$$

$$\text{mIoU} = \frac{1}{2} \times \left(\frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} + \frac{\text{TN}}{\text{TN} + \text{FN} + \text{TP}}\right) \tag{18}$$

where true positives (TPs) denote the number of pixels correctly predicted as cropland, false positives (FPs) represent pixels incorrectly predicted as cropland that are actually noncropland, false negatives (FNs) are the number of pixels incorrectly predicted as noncropland that are actually cropland, and true negatives (TNs) denote pixels correctly predicted as noncropland.

## IV. RESULTS

### A. CFP Mapping Results and Analysis

The CTHBNet model was employed to extract the CFPs across the study areas. The resulting evaluation metrics are summarized in Table IV. In all four regions, CTHBNet

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:08:14 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
ACCURACY EVALUATION OF CFP EXTRACTION RESULTS IN THE FOUR REGIONS OF CHINA

| Dataset | OA (%) | Precision (%) | Recall (%) | F1-score (%) | mIoU (%) |
|---|---|---|---|---|---|
| Dangyang | 95.56 | 94.83 | 94.51 | 94.67 | 91.28 |
| Dezhou | 96.72 | 97.18 | 97.69 | 97.44 | 93.16 |
| Yushu | 92.76 | 94.40 | 95.42 | 94.91 | 88.05 |
| Nanchuan | 98.07 | 85.09 | 80.49 | 82.72 | 84.26 |

achieves OA values exceeding 92.5%, indicating a strong extraction performance. Moreover, the model consistently demonstrates a high performance in the other metrics, with the precision above 85%, recall above 80%, F1-score above 82%, and mIoU above 84%. Among these regions, the best performance is achieved in Dezhou, Shandong, where the cropland parcels are relatively regular in shape with well-defined boundaries, facilitating accurate segmentation by the model. In contrast, the performance in Yushu, Jilin, and Nanchuan, Chongqing, is slightly lower compared to the other two regions. This can be attributed to several factors: In Yushu, Jilin, although the cropland parcels are relatively large, their irregular shapes and poorly defined boundaries significantly affected the accuracy of the extraction results. In Nanchuan, Chongqing, the dominant land type is terraced farming, where the parcels are small and elongated, presenting a major challenge for accurate extraction. Nevertheless, it is important to note that the CTHBNet model still achieved commendable accuracy, demonstrating its robust performance even under these challenging conditions. Notably, although Nanchuan in Chongqing records the highest OA, at 98.07%, it exhibits the lowest scores in the other evaluation metrics, with the precision, recall, F1-score, and mIoU reaching 85.09%, 80.49%, 82.72%, and 84.26%, respectively. This can be attributed to the dominant presence of terraced fields in the region, which are typically narrow and small in area. These characteristics result in a high contrast between foreground and background, contributing to the high OA, but the minimal spacing between adjacent terraces and the frequent adhesion of cropland objects in the imagery compromise the mapping accuracy of individual parcels. In spite of these challenges, CTHBNet maintains a performance above 80% across all the indicators in each region, underscoring its robustness and effectiveness in delineating CFPs in diverse agricultural environments.

Fig. 9 visually illustrates the CFP mapping results across the four regions. The CTHBNet model accurately delineates the parcel boundaries, demonstrating a strong performance in the various landscapes. In Dangyang, Hubei [Fig. 9(a)], the terrain is complex and diverse, comprising plains, hills, and mountainous areas, with both regular strip-shaped fields and irregular parcels. Nevertheless, the model effectively identifies and extracts these varied cropland patterns. In Dezhou, Shandong [Fig. 9(b)], and Yushu, Jilin [Fig. 9(c)], the cropland parcels are relatively regular in shape and larger in size, enabling more straightforward and accurate mapping. In contrast, the cropland in Nanchuan, Chongqing [Fig. 9(d)], is predominantly terraced due to the mountainous topography. These parcels tend to be narrow and small, making the mapping more challenging. In spite of this, CTHBNet effectively addresses these complexities and achieves satisfactory results, demonstrating its adaptability to different cropland morphologies.

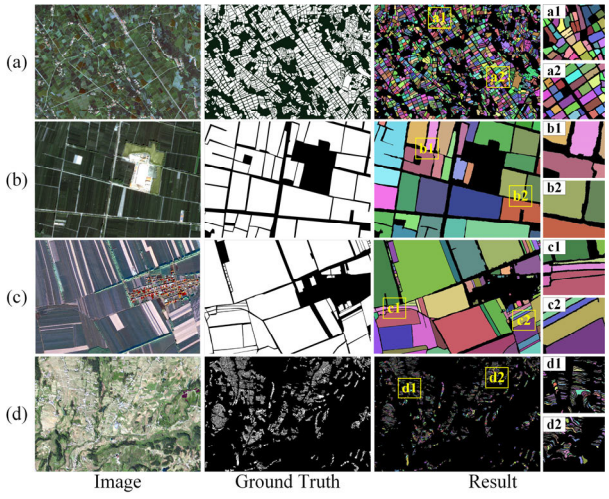



Fig. 9. Results of the CFP mapping in China. (a)–(d) CFP mapping results for Dangyang in Hubei, Dezhou in Shandong, Yushu in Jilin, and Nanchuan in Chongqing, respectively.

TABLE V
ACCURACY EVALUATION OF THE CFP MAPPING RESULTS OF THE DIFFERENT MODELS ON THE DANGYANG DATASET

| Model | OA (%) | Precision (%) | Recall (%) | F1-score (%) | mIoU (%) |
|---|---|---|---|---|---|
| SAM | 77.16 | 72.66 | 72.42 | 72.54 | 62.10 |
| UNet | 87.35 | 84.36 | 85.49 | 84.92 | 77.07 |
| ResUNet | 88.07 | 85.61 | 85.79 | 85.70 | 78.20 |
| DeepLabv3+ | 88.94 | 86.23 | 87.41 | 86.82 | 79.66 |
| PSPNet | 89.55 | 87.07 | 87.98 | 87.53 | 80.66 |
| HRNet | 89.85 | 87.68 | 88.00 | 87.84 | 81.15 |
| SwinUNet | 88.17 | 85.55 | 86.16 | 85.85 | 78.38 |
| UNetFormer | 89.94 | 87.64 | 88.31 | 87.97 | 81.31 |
| CMTFNet | 91.71 | 89.87 | 90.28 | 90.07 | 84.33 |
| PsiNet | 88.50 | 86.71 | 85.51 | 86.11 | 78.87 |
| BsiNet | 88.58 | 87.70 | 84.43 | 86.03 | 78.94 |
| SEANet | 89.22 | 85.35 | 89.48 | 87.36 | 80.19 |
| DBBANet | 92.52 | 91.75 | 91.81 | 91.78 | 86.38 |
| **CTHBNet** | **95.56** | **94.83** | **94.51** | **94.67** | **91.28** |

Note: Bold numbers indicate the highest metric scores across the considered techniques, whereas the underlined numbers represent the second-highest metric scores.

## B. Results of the Comparative Methods

To comprehensively evaluate the CFP mapping performance of the proposed CTHBNet model, we conducted comparative experiments with 13 representative deep-learning models: UNet [50], ResUNet [51], DeepLabv3+ [52], PSPNet [53], HRNet [54], SwinUNet [55], UNetFormer [56], CMTFNet [57], PsiNet [58], BsiNet [43], SEANet [44], DBBANet [11], and SAM [59]. Among the different models, UNet, ResUNet, DeepLabv3+, PSPNet, and HRNet are widely used baseline

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:08:14 UTC from IEEE Xplore. Restrictions apply.

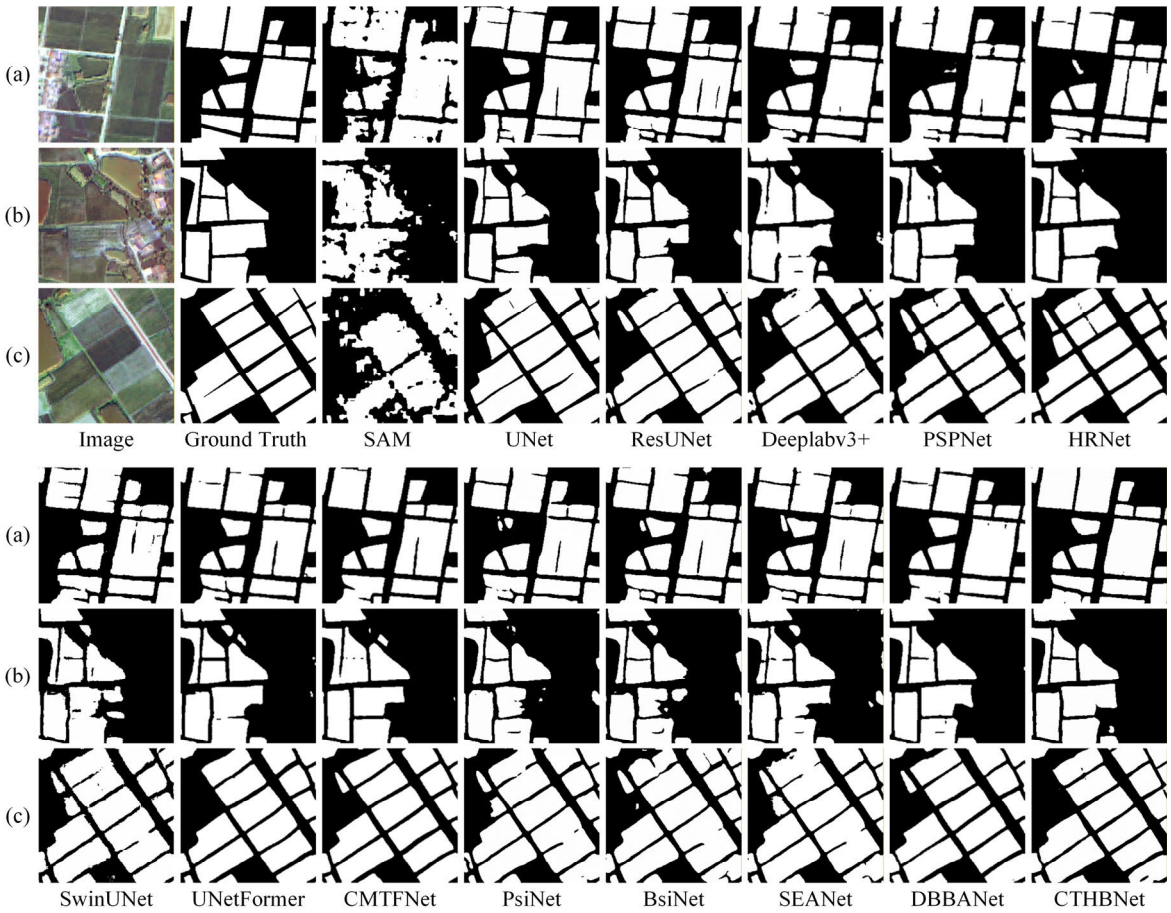



Fig. 10. Results of the model comparison experiments in Dangyang County. (a)–(c) Three detailed display areas within the experimental region.

semantic segmentation networks commonly applied to cropland mapping. SwinUNet is a transformer-based model that replaces all convolutional operations in the original UNet with Swin transformer blocks. UNetFormer and CMTFNet are CNN-transformer hybrid networks that leverage both local and global feature representations. PsiNet is a multitask semantic segmentation model that jointly predicts segmentation masks, boundaries, and distance maps to enhance the segmentation accuracy. BsiNet is another multitask framework specifically designed for CFP mapping from high-resolution satellite imagery, incorporating boundary and distance prediction as auxiliary tasks to improve the performance of the main segmentation task. SEANet is a semantic edge-aware multitask network designed to produce closed and complete agricultural field boundaries in remote sensing imagery. DBBANet is a dual-branch deep network tailored CFP mapping model, which combines a spatial-aware branch and a boundary-aware branch to effectively address shape variability and boundary ambiguity, enabling fine-grained segmentation in high-resolution images. SAM is a foundation model for image segmentation, which provides a strong benchmark for assessing the generalization capability of deep models in large-scale cropland extraction. All the models were trained and evaluated on the same Dangyang dataset, and all experimental parameters were kept consistent. The results are summarized in Table V.

The quantitative results clearly demonstrate that CTHBNet achieves the highest performance across all the evaluation metrics, with each score exceeding 91%. Compared to the second-best model, DBBANet, CTHBNet achieves improvements of 3.04%, 3.08%, 2.70%, 2.89%, and 4.90% in OA, precision, recall, F1-score, and mIoU, respectively. Relative to the lowest-performing model, SAM, CTHBNet exhibits substantial gains of 18.40%, 22.17%, 22.09%, 22.13%, and 29.18% in the same metrics. The second-worst performer is UNet, whose OA, Precision, Recall, F1-score, and mIoU are 8.21%, 10.47%, 9.02%, 9.75%, and 14.21% lower than those of CTHBNet, respectively. The analysis indicates that SAM performs significantly worse than UNet in CFP extraction. This is primarily because SAM is pre-trained on large natural image datasets, which do not adequately capture the spectral, spatial, and semantic characteristics of agricultural landscapes in high-resolution remote sensing imagery. Among the remaining models, CMTFNet and UNetFormer perform slightly poorer than DBBANet but still reach OA values close to 90%, highlighting the effectiveness of integrating global and local features through CNN and transformer fusion. The multitask models, including SEANet, BsiNet, and PsiNet, achieve lower accuracies than the CNN-transformer fusion models but outperform the general-purpose semantic segmentation models. This indicates that

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:08:14 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
ACCURACY EVALUATION OF THE ABLATION EXPERIMENT RESULTS ON THE DANGYANG DATASET

| Number | CTF | HIFD | BGFE | OA (%) | Precision (%) | Recall (%) | F1-score (%) | mIoU (%) |
|---|---|---|---|---|---|---|---|---|
| N1 | | | | 87.35 | 84.36 | 85.49 | 84.92 | 77.07 |
| N2 | √ | | | 89.76 | 88.01 | 87.33 | 87.67 | 80.98 |
| N3 | | √ | | 88.09 | 85.19 | 86.42 | 85.80 | 78.26 |
| N4 | | | √ | 88.15 | 85.96 | 86.51 | 86.23 | 78.61 |
| N5 | √ | √ | | 91.36 | 88.97 | 90.47 | 89.71 | 83.74 |
| N6 | √ | | √ | 90.14 | 87.74 | 88.74 | 88.24 | 81.66 |
| N7 | | √ | √ | 89.34 | 86.22 | 87.14 | 86.68 | 79.34 |
| N8 | √ | √ | √ | 95.56 | 94.83 | 94.51 | 94.67 | 91.28 |

MTL contributes positively to improving the accuracy of CFP mapping.

Fig. 10 presents detailed visual comparisons of the CFP mapping results of the various models. The SAM model exhibits the poorest performance, as field parcels are largely conflated and significant portions of cropland areas remain unrecognized. Among the general semantic segmentation models, UNet, ResUNet, DeepLabv3+, and PSPNet exhibit considerable over-segmentation and parcel adhesion, failing to delineate the parcel boundaries accurately. HRNet, benefiting from multiscale information fusion, alleviates some of the adhesion issues but still struggles in complex areas. For the CNN-transformer hybrid models, CMTFNet and UNetFormer perform well in regions with regular parcels [Fig. 10(c)], but noticeable adhesion issues remain in the irregular regions [Fig. 10(a)–(b)]. Surprisingly, in spite of using multitask collaboration, SEANet, BsiNet, and PsiNet still suffer from significant parcel adhesion and insufficient boundary refinement, which can be attributed to the indistinct boundaries in the specific test areas. DBBANet achieves a better segmentation performance and distinguishes the parcels effectively. However, in regions characterized by irregular shapes and blurred boundaries, a few adhesion issues persist. In contrast, the proposed CTHBNet model demonstrates a superior performance in all scenarios. CTHBNet accurately delineates cropland boundaries across diverse and complex regions, yielding results that are more consistent with the actual spatial distribution of the field parcels.

### C. Ablation Experiment Analysis

To evaluate the effectiveness of each proposed module, we conducted a series of ablation experiments on the Dangyang dataset, as shown in Table VI. Among the different models, number N1 serves as the baseline model, which adopts the classic UNet architecture. For the other ablation groups without the CTF-Module, the encoder structure of UNet was uniformly used to ensure consistency and fairness in the comparison.

The quantitative results clearly demonstrate that each of the proposed modules contributes positively to the CFP mapping. Among the different modules, the CTF-Module shows the most significant improvement. After incorporating this module, the model's OA, precision, recall, F1-score, and mIoU are increased by 2.41%, 3.65%, 1.84%, 2.75%, and 3.91%, respectively. This highlights the substantial effectiveness of the CTF-Module and the advantage of combining global and local information. The next most effective modules are the BGFE-Module and HIF-Module. After adding these modules individually, the model's OA is improved by 0.80% and 0.74%, the F1-score is improved by 1.26% and 0.88%, and the mIoU is improved by 1.54% and 1.19%, respectively. Furthermore, combining any two of the modules leads to a better performance than using any single module alone, and integrating all three modules achieves the best results. This indicates that the designed modules can work synergistically to enhance CFP mapping. To further illustrate the ablation results, error maps were generated, as shown in Fig. 11.

After integrating all three proposed modules [Fig. 11(N8)], the occurrence of misclassification and omission errors is significantly reduced, resulting in extraction outcomes that more closely reflect real-world cropland patterns. This highlights the superior performance achieved through the joint contribution of these modules in CFP mapping. Compared to the results in Fig. 11(N1–N4), both misclassified and omitted regions are markedly diminished, and the continuity and completeness of parcel boundaries are visibly improved, demonstrating the positive impact of each module. Moreover, after incorporating the CTF module [Fig. 11(N2, N5, N6, and N8)], the model exhibits excellent performance in identifying large-scale cropland parcels, mainly due to the effective integration of global and local spatial information, which allows the model to better capture the context of large areas and minimize errors in these regions. In addition, as shown in Fig. 11(N4), the introduction of the BGFE-Module effectively reduces segmentation adhesion, particularly for elongated parcels with ambiguous boundaries, thereby improving boundary clarity and resulting in more accurate and complete mapping outcomes.

## V. DISCUSSION

### A. Explanation of the Effectiveness of the CTF-Module

As demonstrated in the previous analysis, the proposed CTF-Module significantly enhances CFP mapping performance by effectively integrating global and local information. However, the key to the success of the CTF-Module lies in how these two types of information are fused. To evaluate the effectiveness of the fusion approach, we compared the CTF-Module with five alternative feature fusion strategies: the feature aggregation module (FAM) [60], the residual

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:08:14 UTC from IEEE Xplore. Restrictions apply.

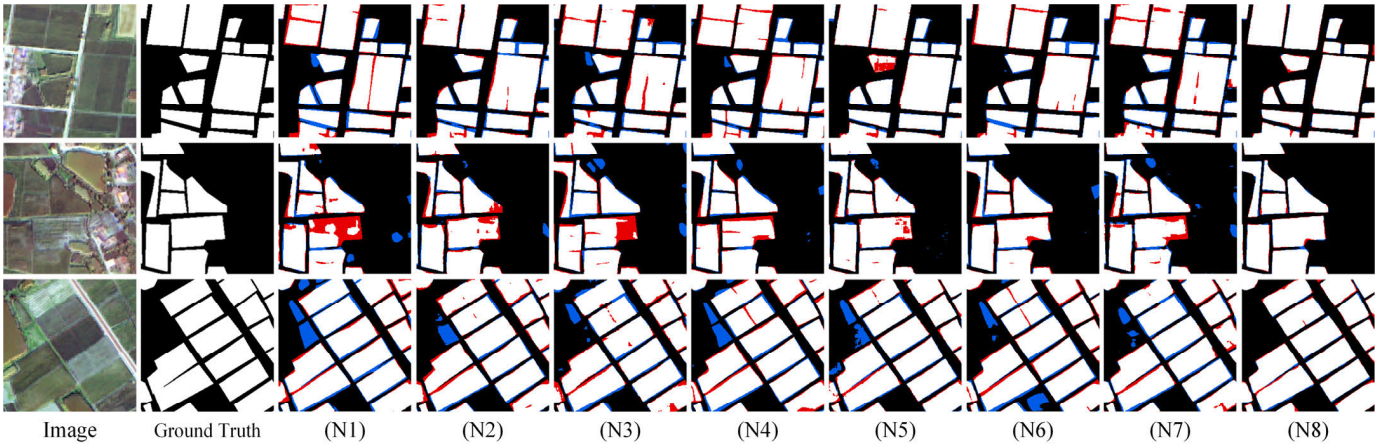



Fig. 11. Error analysis maps of the ablation experiment results. Red indicates omission errors, and blue indicates commission errors. N1–N8 correspond to the "number" entries in Table VI.

TABLE VII
ACCURACY EVALUATION OF THE CFP MAPPING RESULTS WITH DIFFERENT CNN-TRANSFORMER FUSION STRATEGIES ON THE DANGYANG DATASET

| Strategy | OA (%) | Precision (%) | Recall (%) | F1-score (%) | mIoU (%) |
|---|---|---|---|---|---|
| CTF (proposed) | 95.56 | 94.83 | 94.51 | 94.67 | 91.28 |
| CNN | 88.07 | 85.61 | 85.79 | 85.70 | 78.20 |
| Transformer | 91.69 | 89.69 | 90.44 | 90.07 | 84.29 |
| FAM | 89.95 | 87.59 | 88.39 | 87.99 | 81.32 |
| RAM | 91.73 | 90.01 | 90.16 | 90.08 | 84.36 |
| LAFM | 90.72 | 88.72 | 89.05 | 88.88 | 82.62 |
| Concat | 90.80 | 88.41 | 89.68 | 89.04 | 82.78 |
| Addition | 92.27 | 90.11 | 91.49 | 90.80 | 85.33 |

attention module (RAM) [61], the local-aware fusion module (LAFM) [62], simple concatenation (Concat), and element-wise Addition (Addition). Furthermore, to assess the individual contributions of the global and local information, we also included models that rely solely on CNN or transformer branches for comparison. All the experiments were conducted on the Dangyang dataset. The results are summarized in Table VII.

After dual-branch fusion, the model outperforms both single-branch models across all the evaluation metrics, with the transformer branch performing better than the CNN branch. Compared to the CNN branch, the fusion model achieves improvements of 7.49%, 9.22%, 8.72%, 8.97%, and 13.08% in OA, precision, recall, F-score, and mIoU, respectively. Compared to the transformer branch, these metrics are improved by 3.87%, 5.14%, 4.07%, 4.60%, and 6.99%, respectively. This is because the transformer model, through its self-attention mechanism, captures the dependencies among all the pixels in the imagery, enabling effective modeling of the global contextual information, which is particularly important for farmland plots with fuzzy boundaries or irregular shapes. In contrast, CNNs have limited receptive fields and struggle with long-range dependencies, which constrains their ability to extract features from complex regions. By integrating both branches, the model leverages their complementary strengths and achieves an improved OA.

In the comparison with five global-local information fusion strategies, the proposed strategy achieves the best performance across all the evaluation metrics, demonstrating the effectiveness of the proposed fusion strategy in the model design. Compared to the second-best strategy, Addition, the model improves the OA, precision, recall, F-score, and mIoU by 3.29%, 4.72%, 3.02%, 3.87%, and 5.95%, respectively. Compared to the weakest strategy, FAM, the improvements reach 5.61%, 7.24%, 6.12%, 6.68%, and 9.96%, respectively. The core of the proposed strategy is a stacking operation, which enables more thorough integration of information. In addition, to emphasize local features such as texture, the local importance of the features is first enhanced in the CNN branch, feature stacking is then applied, followed by channel- and spatial-wise attention mechanisms to further strengthen the feature representations, ultimately achieving the effective fusion of global and local information.

To further illustrate the effectiveness of the proposed fusion strategy, we visualized the model's attention using heatmaps (Fig. 12). The proposed strategy directly focuses on the cropland regions, clearly distinguishing them from noncropland areas. The RAM strategy mainly concentrates on the cropland boundaries to achieve delineation, while the FAM strategy fails to consistently focus on cropland regions. Although the LAFM and Concat strategies can attend to cropland areas, they exhibit confusion with noncropland regions, leading to adhesion artifacts in the results. The second-best strategy, Addition, demonstrates accurate focus on cropland, but it also pays attention to other objects, causing misclassification. In summary, the heatmap analysis further supports the effectiveness of the proposed fusion strategy, which allows the model to concentrate on cropland areas while clearly distinguishing boundaries from surrounding noncropland, leading to complete and more accurate CFP mapping.

### B. Generalization Capability of CTHBNet in Global Regions

Evaluating the generalization and transferability of a model is crucial to ensuring its stable performance in complex

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:08:14 UTC from IEEE Xplore. Restrictions apply.

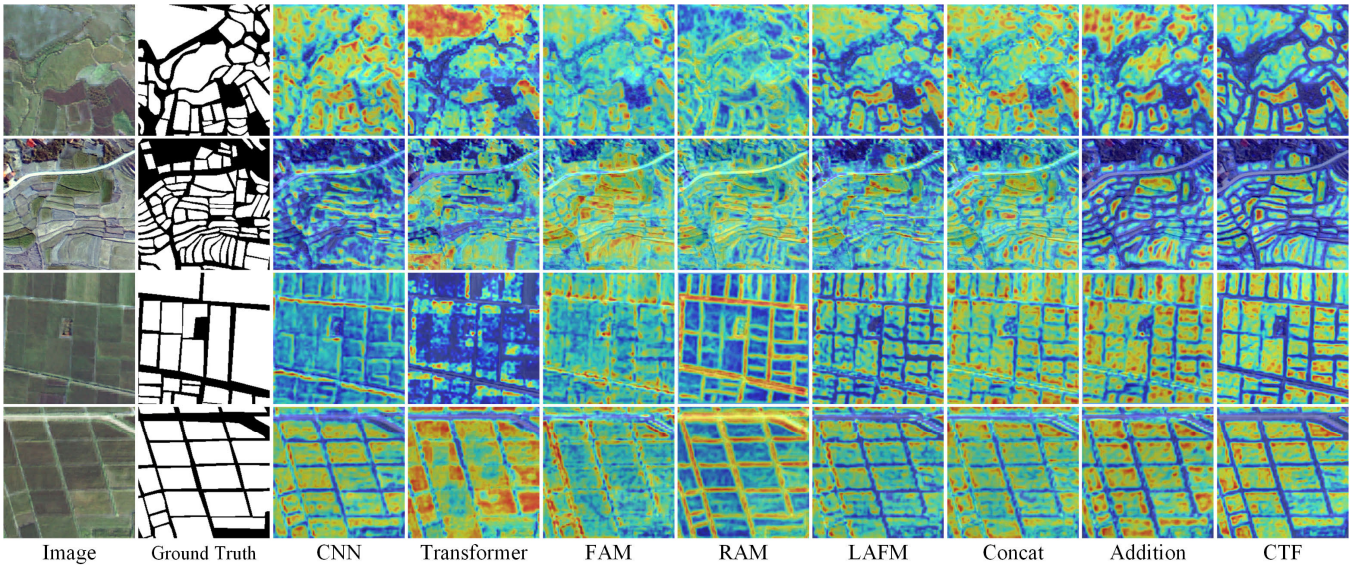



Fig. 12. Heatmaps of the model attention regions under different CNN-transformer fusion strategies. The color transitions from blue to red indicate increasing levels of model attention.

TABLE VIII

ACCURACY EVALUATION OF THE MIGRATION EXPERIMENT RESULTS USING THE CTHBNET MODEL ON A GLOBAL SCALE

| Region | OA (%) | Precision (%) | Recall (%) | F1-score (%) | mIoU (%) |
|---|---|---|---|---|---|
| Europe (T1) | 96.46 | 97.43 | 97.98 | 97.71 | 90.54 |
| Africa (T2) | 96.23 | 97.23 | 97.72 | 97.47 | 90.63 |
| Oceania (T3) | 95.42 | 96.94 | 97.11 | 97.03 | 88.08 |
| North America (T4) | 97.49 | 98.43 | 98.56 | 98.50 | 91.46 |
| South America (T5) | 94.27 | 95.93 | 96.37 | 96.15 | 86.74 |

real-world scenarios. As demonstrated in the previous experiments, the CTHBNet model achieved promising results in CFP mapping across typical regions in China. However, due to the significant differences in agricultural practices, natural environments, and remote sensing image characteristics worldwide, the applicability of CTHBNet to other regions remains uncertain. To address this, we selected representative agricultural areas from Europe, Africa, Oceania, North America, and South America to comprehensively assess the cross-regional generalization and transfer capability of the CTHBNet model. The evaluation metrics for the experimental results are presented in Table VIII.

The migration experiments conducted across the five additional regions demonstrate that the CTHBNet model achieves OAs above 94%, F1-score values exceeding 96%, and mIoU values above 86%, indicating the strong transferability and generalization capability. Further comparison reveals that the model performs best in the United States (North America), with the OA, precision, recall, F1-score, and mIoU reaching 97.49%, 98.43%, 98.56%, 98.50%, and 91.46%, respectively. In contrast, the weakest performance is observed in Chile (South America), with corresponding values of 94.27%, 95.95%, 96.37%, 96.15%, and 86.74%. Among the remaining regions, the performance ranks from high to low as follows: Australia (Oceania), the Netherlands (Europe), and South Africa (Africa).

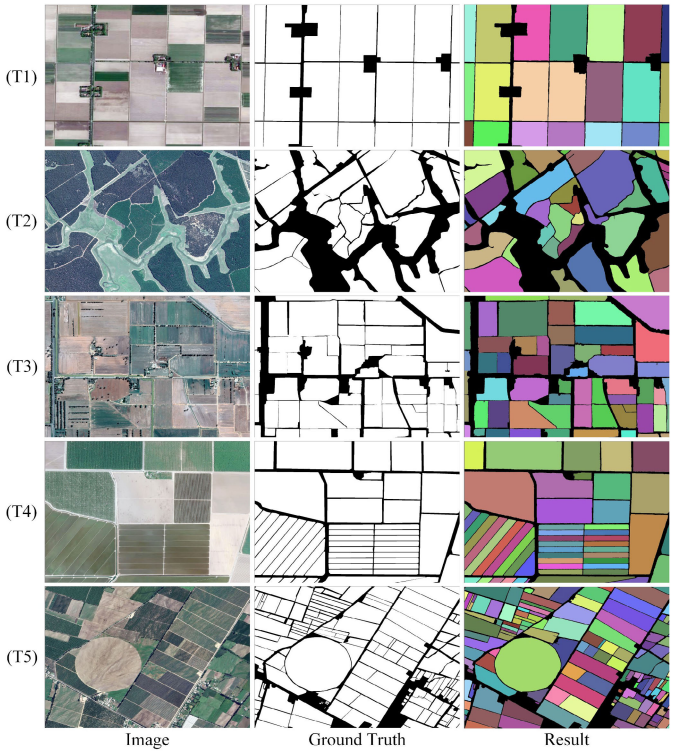



Fig. 13. Results of the CFP mapping in global regions. T1–T5 correspond to the CFP mapping results from selected regions in the Netherlands (Europe), South Africa (Africa), Australia (Oceania), the United States (North America), and Chile (South America), respectively.

Fig. 13 visually illustrates the CFP mapping results of the CTHBNet model across the different global regions. As shown, in the Netherlands (T1) and the United States (T4), where the cropland parcels are regular in shape and have

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:08:14 UTC from IEEE Xplore. Restrictions apply.

clearly defined boundaries, the model achieves complete and accurate mapping results. In Chile (T5), where both regular and fragmented parcels are present, CTHBNet is still able to effectively extract and accurately delineate the cropland boundaries. In Australia (T3), although the cropland parcels are relatively regular and large in area, their boundaries are more ambiguous, and the background is more complex, posing challenges to accurate mapping. In spite of some instances of adhesion or omission, the model maintains a strong boundary segmentation capability and consistent recognition performance. In South Africa (T2), where the cropland parcels vary greatly in shape and size and share spectral and textural similarities with the surrounding noncropland classes, CTHBNet is still capable of distinguishing cropland from noncropland areas and preserving the integrity of the parcel boundaries. Overall, CTHBNet demonstrates a strong generalization ability and stable mapping performance across diverse geographical regions and agricultural practices, further validating its potential for wide application in multiregional and multiscenario contexts.

To further demonstrate the performance of the CTHBNet model across different regions, we conducted comparative experiments with several representative models, including DBBANet, BsiNet, CMTFNet, UNetFormer, DeepLabv3+, and UNet. These models cover a range of categories: state-of-the-art CFP mapping methods, hybrid CNN-transformer architectures, and general-purpose semantic segmentation models. The results are presented in Table IX.

Through the quantitative analysis, it can be seen that the CTHBNet model achieves the best performance across all the regions, highlighting its superior capabilities. CMTFNet ranks second in regions T1 and T2, while DBBANet performs second-best in regions T3, T4, and T5. The UNet model consistently performs the worst in all the regions, with the OA lagging by 0.92%, 0.93%, 1.76%, 0.51%, and 1.26%, respectively. Its precision falls behind by 0.68%, 0.51%, 1.29%, 0.13%, and 1.31%, recall by 0.49%, 0.74%, 0.98%, 0.48%, and 0.61%, F1-score by 0.59%, 0.62%, 1.14%, 0.31%, and 0.96%, and mIoU by 2.28%, 2.13%, 4.18%, 1.55%, and 2.66%, respectively. Among the remaining models, the performance ranking from best to worst is UNetFormer, BsiNet, and DeepLabv3+. Overall, CMTFNet and UNetFormer are ranked among the top-performing models, indicating that the effective integration of transformer and CNN architectures can significantly improve CFP mapping, which is consistent with the motivation behind the design of the proposed model. In addition, in region T4, the accuracy differences among all the models are relatively small, mainly because the cropland parcels in this area are large and regularly shaped, allowing even the general-purpose semantic segmentation models to achieve satisfactory results. In contrast, the specialized CFP mapping models are designed to handle more challenging parcel types, such as small or elongated plots, which require detailed feature learning. Therefore, the advantages of CTHBNet are more pronounced in regions such as T3 and T5.

Fig. 14 illustrates the detailed experimental results of the comparison models across the five regions. In region T1_1, in spite of the extremely narrow spacing between adjacent cropland parcels, the proposed model accurately identifies and delineates the boundaries, while the other models fail to make these distinctions. In region T1_2, the spectral characteristics of some cropland parcels differ significantly from the surrounding land-cover types, leading to omission errors in several of the models. In contrast, the proposed model successfully extracts these parcels in full, demonstrating its superior discrimination and robustness.

In region T2_1, the red oval area represents cropland that was mistakenly omitted during the annotation, while the yellow oval area was incorrectly labeled as cropland. A comparison reveals that such annotation errors have a minimal impact on the proposed model, which still accurately identifies the actual cropland boundaries. Although these labeling inaccuracies can lead to slightly lower evaluation scores, the extracted results align more closely with the real-world conditions. In region T2_2, the red and blue ovals highlight the differences in model performance. The proposed

TABLE IX
ACCURACY EVALUATION OF THE CFP MAPPING RESULTS UNDER THE DIFFERENT MODELS ON A GLOBAL SCALE

| Region | Model | OA (%) | Precision (%) | Recall (%) | F1-score (%) | mIoU (%) |
|---|---|---|---|---|---|---|
| T1 | **CTHBNet** | **96.46** | **97.43** | **97.98** | **97.71** | **90.54** |
| | DBBANet | 96.25 | 97.34 | 97.81 | 97.57 | 90.03 |
| | BsiNet | 95.69 | 96.82 | 97.61 | 97.21 | 88.61 |
| | CMTFNet | 96.36 | 97.35 | 97.85 | 97.60 | 90.28 |
| | UNetFormer | 96.14 | 97.18 | 97.83 | 97.50 | 89.74 |
| | DeepLabv3+ | 95.98 | 97.02 | 97.78 | 97.40 | 89.32 |
| | UNet | 95.54 | 96.75 | 97.49 | 97.12 | 88.26 |
| T2 | **CTHBNet** | **96.23** | **97.23** | **97.72** | **97.47** | **90.63** |
| | DBBANet | 95.88 | 97.07 | 97.41 | 97.24 | 89.81 |
| | BsiNet | 95.68 | 96.93 | 97.27 | 97.10 | 89.35 |
| | CMTFNet | 96.05 | 97.20 | 97.50 | 97.35 | 90.21 |
| | UNetFormer | 95.76 | 96.95 | 97.36 | 97.16 | 89.53 |
| | DeepLabv3+ | 95.52 | 96.78 | 97.22 | 97.00 | 88.99 |
| | UNet | 95.30 | 96.72 | 96.98 | 96.85 | 88.50 |
| T3 | **CTHBNet** | **95.42** | **96.94** | **97.11** | **97.03** | **88.08** |
| | DBBANet | 95.15 | 96.75 | 96.95 | 96.85 | 87.41 |
| | BsiNet | 93.93 | 95.85 | 96.27 | 96.06 | 84.52 |
| | CMTFNet | 94.79 | 96.37 | 96.88 | 96.62 | 86.52 |
| | UNetFormer | 94.25 | 96.35 | 96.16 | 96.26 | 85.36 |
| | DeepLabv3+ | 94.25 | 95.79 | 96.77 | 96.28 | 85.17 |
| | UNet | 93.66 | 95.65 | 96.13 | 95.89 | 83.90 |
| T4 | **CTHBNet** | **97.49** | **98.43** | **98.56** | **98.50** | **91.46** |
| | DBBANet | 97.48 | 98.43 | 98.55 | 98.49 | 91.41 |
| | BsiNet | 97.33 | 98.41 | 98.39 | 98.40 | 90.98 |
| | CMTFNet | 97.38 | 98.39 | 98.47 | 98.43 | 91.11 |
| | UNetFormer | 97.24 | 98.40 | 98.29 | 98.35 | 90.71 |
| | DeepLabv3+ | 97.21 | 98.21 | 98.45 | 98.33 | 90.54 |
| | UNet | 96.98 | 98.30 | 98.08 | 98.19 | 89.91 |
| T5 | **CTHBNet** | **94.27** | **95.93** | **96.37** | **96.15** | **86.74** |
| | DBBANet | 94.11 | 95.81 | 96.04 | 95.93 | 86.49 |
| | BsiNet | 93.31 | 94.82 | 95.97 | 95.38 | 84.70 |
| | CMTFNet | 93.88 | 95.63 | 95.90 | 95.77 | 85.99 |
| | UNetFormer | 93.39 | 94.71 | 96.22 | 95.46 | 84.84 |
| | DeepLabv3+ | 93.15 | 94.98 | 95.57 | 95.27 | 84.44 |
| | UNet | 93.01 | 94.62 | 95.76 | 95.19 | 84.08 |

Note: Bold numbers indicate the highest metric scores across the considered techniques, whereas the underlined numbers represent the second-highest metric scores.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:08:14 UTC from IEEE Xplore. Restrictions apply.

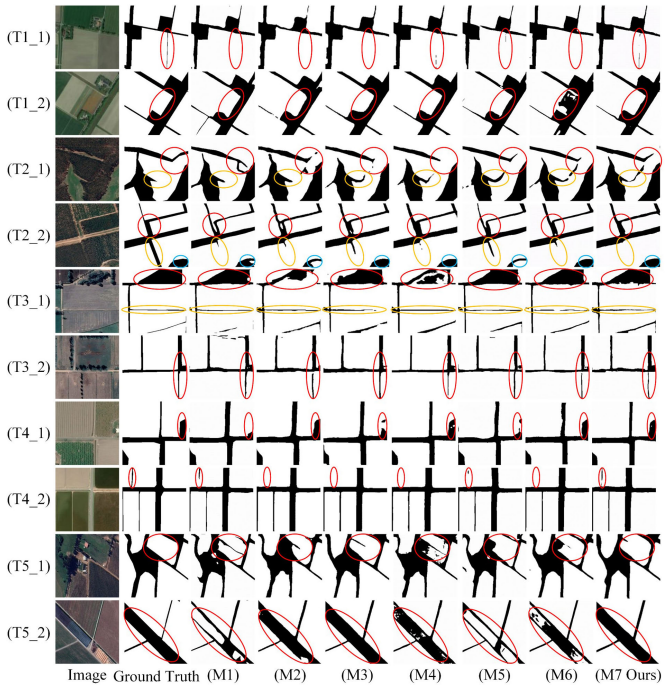



Fig. 14. Detailed display of the CFP mapping results in global regions with the different models. T1–T5 correspond to the CFP mapping results from selected regions in the Netherlands (Europe), South Africa (Africa), Australia (Oceania), the United States (North America), and Chile (South America), respectively. M1–M7 refer to the CFP mapping results of the UNet, DeepLabV3+, UNetFormer, CMTFNet, BsiNet, DBBANet, and CTHBNet, respectively.

model accurately identifies the cropland parcels, whereas the other models exhibit varying degrees of misclassification. In addition, the yellow oval area is in fact cropland, although it was not labeled as such in the ground truth. Nonetheless, the proposed model correctly extracts the area, demonstrating strong generalization and robustness to labeling errors.

In T3_1, DeepLabv3+, UNetFormer, and CMTFNet fail to distinguish the noncropland areas marked by the red oval, resulting in FPs. Meanwhile, UNet, BsiNet, and DBBANet are unable to correctly detect the narrow field ridges in the yellow oval, causing misclassifications. In contrast, the proposed model not only distinguishes cropland from noncropland but also reduces edge adhesion, producing more complete mapping results. In T3_2, although DBBANet and the other comparison models are able to identify some adjacent cropland parcels, they fail to properly delineate the ridges, leading to boundary merging. The proposed model, however, can accurately extract cropland parcels with clear boundaries, yielding better overall results.

In T4_1, UNet, UNetFormer, and BsiNet mistakenly segment the small buildings located within the cropland areas, resulting in false detections. The proposed model can accurately identify such small structures and delineate the parcel boundaries correctly. In the subregion of T4_2, the proposed model again successfully detects the very narrow gaps between adjacent cropland parcels, where the other models fail to make a distinction.

In T5_1, the other models exhibit noticeable undersegmentation, particularly in the red oval area. The proposed model, in contrast, provides a more faithful representation of the cropland parcels with complete boundary delineation. In T5_2, interference from spectrally similar features or the complex background leads most of the models to extract noncropland regions, resulting in significant misclassification. The proposed model remains unaffected by this interference and correctly differentiates the similar land features from actual cropland.

In summary, the proposed model consistently demonstrates a superior performance across all five regions. It effectively distinguishes the background and the visually similar land-cover types, reduces misclassification, and even compensates for the annotation errors in the dataset. Furthermore, it delivers more complete CFP mapping results compared to the other baseline models, thereby showcasing strong transferability and generalization capabilities.

### C. Evaluation of CTHBNet on Public Benchmark Datasets

To date, there has been extensive research on CFP mapping, and several benchmark datasets have been built by scholars, such as the fine-grained farmland dataset (FGFD) [11], the Ai4Boundary dataset [63], and the Denmark dataset. The FGFD dataset consists of Google Earth imagery with a spatial resolution of 0.3 m; the Ai4Boundary dataset comprises orthophotos with a 1-m resolution; and the Denmark dataset is based on Sentinel-2 imagery with a resolution of 10 m.

The FGFD dataset covers diverse cropland landscapes across seven major geographical terrains in China. Specifically, the Tibetan Plateau accounts for the largest proportion of the dataset (32.9%), followed by the Yunnan-Guizhou Plateau (23.0%), the North China Plain (8.6%), the Northeast Plain (8.3%), the Middle-Lower Yangtze River Plain (9.2%), the Southeast Hills (8.6%), and the Loess Plateau (9.4%). The Ai4Boundary dataset includes cropland data from various European countries such as Austria, Spain, France, Luxembourg, the Netherlands, Slovenia, and Sweden. The Denmark dataset focuses on the eastern region of Denmark, which is predominantly flat and consists mainly of plains. The cropland parcels in this region are mostly small to medium in size and are scattered due to urbanization. In addition, some parcels are used for mixed farming, including both crop cultivation and livestock. In summary, these datasets collectively represent a wide range of agricultural landscapes and use different types of remote sensing imagery. Conducting experiments on these datasets not only allows for the evaluation of the model's performance across diverse scenarios but also helps verify the model's adaptability to varying datasets and remote sensing sensors. Therefore, we conducted experiments using these datasets. For each benchmark dataset, the samples were divided into training, validation, and testing sets, containing 4000, 500, and 500 pairs, respectively. The results are listed in Table X.

In the experiments conducted on the above three datasets, the CTHBNet model achieves an OA above 88%, an F1-score above 83%, and an mIoU close to 80%, demonstrating good adaptability across the different datasets. The best performance of CTHBNet can be observed on the FGFD dataset, with OA, precision, recall, F1-score, and mIoU values of 90.63%,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:08:14 UTC from IEEE Xplore. Restrictions apply.

TABLE X
ACCURACY EVALUATION OF THE CFP MAPPING RESULTS ON THE BENCHMARK DATASETS

| Dataset | OA (%) | Precision (%) | Recall (%) | F1-score (%) | mIoU (%) |
|---|---|---|---|---|---|
| FGFD | 90.63 | 91.32 | 95.89 | 93.55 | 79.47 |
| Ai4boundary | 88.71 | 87.60 | 83.95 | 85.74 | 78.75 |
| Denmark | 88.34 | 82.43 | 84.52 | 83.46 | 77.84 |

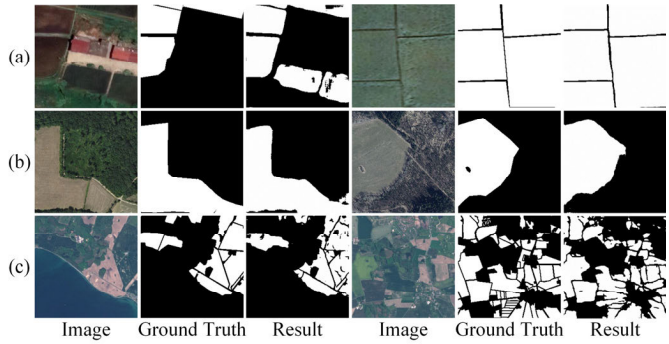



Fig. 15. CFP mapping results of CTHBNet on the different benchmark datasets. (a)–(c) FGFD, Ai4boundary, and Denmark datasets, respectively.

91.32%, 95.89%, 93.55%, and 79.47%, respectively, followed by the Ai4Boundary and Denmark datasets. The spatial resolution of the imagery decreases progressively from FGFD to Ai4Boundary to Denmark, and the cropland in the Denmark dataset is more fragmented, which contributes to the relatively low performance on the Denmark dataset. Nevertheless, the model's precision, recall, and F1-score all remain above 80%, indicating that CTHBNet performs well across these datasets.

Fig. 15 illustrates the detailed experimental results of the CTHBNet model on the various benchmark datasets. For the FGFD [Fig. 15(a)] and Ai4boundary [Fig. 15(b)] datasets, due to their high spatial resolution, the extracted cropland parcel boundaries closely match the actual field shapes. In addition, CTHBNet is able to identify some cropland parcels that were not annotated in the ground truth, demonstrating its strong recognition capability. In contrast, the Denmark dataset [Fig. 15(c)] has a lower spatial resolution of 10 m, and the reference labels contain noticeable jagged edges. As a result, the predicted boundaries from CTHBNet also exhibit jaggedness, and there are some instances of boundary adhesion. Nevertheless, the overall mapping results remain consistent with the actual cropland distribution, reflecting the model's adaptability and robustness across varying data qualities.

To further validate the superiority of the proposed model, comparative experiments with several baseline models were conducted on the three public benchmark datasets. Previous model comparisons indicated that SAM, ResUNet, and PSPNet performed poorly in CFP extraction. Considering the extraction accuracy, functional characteristics, and model types, we ultimately selected DBBANet, BsiNet, CMTFNet, UNetFormer, UNet, DeepLabv3+, and the proposed CTHBNet for detailed experimental evaluation. The quantitative results are summarized in Table XI. Across all three datasets, CTHBNet consistently achieves the best performance, while DBBANet ranks second and UNet shows the poorest results in all cases. On the FGFD dataset, compared to DBBANet, CTHBNet achieves improvements of 0.09%, 0.04%, 0.13%, 0.11%, and 0.12% in OA, precision, recall, F1-score, and mIoU, respectively. Compared to UNet, the improvements are 2.76%, 1.70%, 2.28%, 1.98%, and 5.04%, respectively. On the Ai4boundary dataset, CTHBNet outperforms DBBANet by 0.39%, 0.48%, 0.78%, 0.64%, and 0.79% in OA, precision, recall, F1-score, and mIoU, respectively, and surpasses UNet by 2.64%, 3.70%, 4.18%, 3.96%, and 4.30%, respectively. On the Denmark dataset, CTHBNet also leads, with improvements over DBBANet of 0.58%, 0.76%, 0.33%, 0.55%, and 0.86%, respectively, and improvements over UNet of 4.85%, 7.07%, 8.23%, 7.63%, and 8.44%, respectively. In summary, although the performance gap between CTHBNet and DBBANet is relatively small, CTHBNet consistently achieves superior results across all the datasets. This further confirms the effectiveness of the proposed CTHBNet model in CFP mapping.

### D. Limitations and Future Work

Table XII summarizes the parameter counts and computational efficiency of the compared models. In terms of computational efficiency, the UNet model achieves the highest FPS, at 224.40, while SEANet has the lowest FPS, at only 21.75, due to the extensive feature fusion. HRNet also has a low FPS of 21.85 because it processes multiple branches and larger feature maps. The CTHBNet model, which combines CNN and transformer architectures, involves a large amount of computation, resulting in an FPS of 35.61, placing it in the mid-to-low range. Regarding FLOPs, SEANet has the highest, at 208.14G, UNetFormer has the lowest, at 6.04G, and CTHBNet ranks in the middle, at 23.69G. In terms of parameter count, SAM is the largest with 90.21 million parameters, followed by CTHBNet with 74.03 million, while UNet is the smallest with only 7.85 million. Although CTHBNet has fewer parameters than the foundation model SAM, its parameter count is still much higher than that of most existing mainstream CFP extraction models. In spite of achieving satisfactory accuracy, CTHBNet exhibits relatively high computational complexity and lower inference efficiency, which imposes certain demands on computational resources. Therefore, future work should focus on reducing the parameter count and computational complexity of the CTHBNet model to improve its efficiency. For example, techniques such as model pruning or knowledge distillation could be applied to compress the model, thereby improving the inference speed and resource use while maintaining the accuracy.

In this study, we only investigated the effectiveness of the CTHBNet model for CFP mapping based on high-resolution optical data. We acknowledge that the performance of parcel delineation algorithms is closely related to sensor characteristics, particularly spatial resolution and temporal revisit frequency. Different sensors may require specific design considerations. Many regions worldwide are frequently affected by cloudy and rainy weather, which limits the availability of optical imagery and the model's ability to extract cropland information in those areas. Therefore, future research could explore the integration of active remote sensing data, such as

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:08:14 UTC from IEEE Xplore. Restrictions apply.

TABLE XI
ACCURACY EVALUATION OF THE CFP MAPPING RESULTS UNDER THE DIFFERENT MODELS ON THE BENCHMARK DATASETS

| Dataset | Model | OA (%) | Precision (%) | Recall (%) | F1-score (%) | mIoU (%) |
|---|---|---|---|---|---|---|
| FGFD | **CTHBNet** | **90.63** | **91.32** | **95.89** | **93.55** | **79.47** |
| | DBBANet | 90.59 | 91.28 | 95.76 | 93.47 | 79.35 |
| | BsiNet | 88.39 | 90.36 | 93.48 | 91.89 | 75.52 |
| | CMTFNet | 89.58 | 90.93 | 94.64 | 92.75 | 77.63 |
| | UNetFormer | 89.41 | 90.15 | 95.37 | 92.69 | 77.06 |
| | DeepLabv3+ | 88.63 | 90.11 | 94.19 | 92.10 | 75.81 |
| | UNet | 87.87 | 89.62 | 93.61 | 91.57 | 74.43 |
| Ai4boundary | **CTHBNet** | **88.71** | **87.60** | **83.95** | **85.74** | **78.75** |
| | DBBANet | 88.32 | 87.12 | 83.17 | 85.10 | 77.96 |
| | BsiNet | 86.46 | 83.92 | 80.97 | 82.42 | 75.13 |
| | CMTFNet | 87.52 | 85.33 | 80.77 | 82.99 | 75.97 |
| | UNetFormer | 86.41 | 83.95 | 80.76 | 82.33 | 75.04 |
| | DeepLabv3+ | 86.30 | 83.71 | 80.77 | 82.21 | 74.88 |
| | UNet | 86.07 | 83.90 | 79.77 | 81.78 | 74.45 |
| Denmark | **CTHBNet** | **88.34** | **82.43** | **84.52** | **83.46** | **77.84** |
| | DBBANet | 87.76 | 81.67 | 84.19 | 82.91 | 76.98 |
| | BsiNet | 85.28 | 78.17 | 78.55 | 78.36 | 72.17 |
| | CMTFNet | 86.15 | 80.08 | 78.75 | 79.41 | 73.47 |
| | UNetFormer | 86.47 | 79.55 | 80.92 | 80.23 | 74.17 |
| | DeepLabv3+ | 85.29 | 77.09 | 81.73 | 79.34 | 72.89 |
| | UNet | 83.49 | 75.36 | 76.29 | 75.83 | 69.40 |

Note: Bold numbers indicate the highest metric scores across the considered techniques, whereas the underlined numbers represent the second-highest metric scores.

TABLE XII
COMPUTATIONAL PERFORMANCE OF THE DIFFERENT MODELS

| Model | FPS ↑ | FLOPs (G) ↓ | Params (M) ↓ |
|---|---|---|---|
| SAM | 49.24 | 32.27 | 90.2083 |
| UNet | 224.40 | 13.27 | 7.8466 |
| ResUNet | 128.32 | 23.46 | 69.3056 |
| DeepLabv3+ | 112.67 | 11.72 | 42.3970 |
| PSPNet | 48.44 | 44.73 | 46.5849 |
| HRNet | 21.82 | 23.46 | 65.8464 |
| SwinUNet | 19.34 | 148.78 | 43.4384 |
| UNetFormer | 97.39 | 6.04 | 24.1980 |
| CMTFNet | 69.93 | 8.57 | 30.0696 |
| PsiNet | 97.94 | 32.60 | 7.8467 |
| BsiNet | 122.84 | 24.84 | 7.8467 |
| SEANet | 21.75 | 208.14 | 28.7525 |
| DBBANet | 80.55 | 16.26 | 33.6464 |
| CTHBNet | 35.61 | 23.69 | 74.0326 |

Note: FPS indicates the inference speed (frames per second); FLOPs denotes the computational complexity (in G); Params represents the number of trainable parameters (in M). ↑ indicates that a higher value is better, while ↓ indicates that a lower value is better.

synthetic aperture radar (SAR), to enhance the adaptability and robustness of CTHBNet across diverse sensors and environmental conditions.

Additionally, CFPs exhibit temporal dynamics related to crop growth, but this study did not investigate the sensitivity of the CTHBNet model to different time periods. Future work could leverage multitemporal remote sensing images to assess the model's performance over time and explore spatiotemporal feature fusion strategies to further improve the accuracy and stability of CFP mapping. Collectively, these efforts could contribute to developing a more unified and adaptable framework for agricultural parcel delineation that aligns algorithm design with sensor-specific spatial and temporal properties.

## VI. CONCLUSION

In this article, we have proposed the novel CTHBNet model, which is capable of accurately mapping CFPs from high-resolution remote sensing imagery. The CTHBNet model is based on an encoder-decoder architecture, where the encoder employs bottleneck and Swin transformer blocks as backbones. The effective CTF-Module is designed to integrate comprehensive multiscale parcel information, improving the mapping accuracy and preventing the loss of small and elongated CFPs. In the decoder, the HIF-Module is introduced to progressively restore spatial details, while the BGFE-Module, leveraging Sobel gradients and global attention, refines the boundary representations and mitigates parcel adhesion, omission, and misclassification. In addition, an MTL strategy is used to jointly optimize the parcel extent, boundary, and distance representations to improve the segmentation precision and robustness. Extensive experiments using GF-2 imagery were conducted in four diverse agricultural landscapes in China: Dangyang (Hubei), Dezhou (Shandong), Yushu (Jilin), and Nanchuan (Chongqing). The OA exceeded 92%, and the mIoU exceeded 84% in these areas, indicating that CTHBNet can effectively extract CFPs while maintaining a

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:08:14 UTC from IEEE Xplore. Restrictions apply.

high spatial integrity. On the Dangyang dataset, comparative experiments against general semantic segmentation models, CNN-transformer hybrid segmentation models, multitask segmentation models, vision-language model (SAM), and the existing mainstream CFP mapping methods demonstrated that CTHBNet achieved the highest scores across all the metrics, further highlighting its superior performance. Ablation experiments also confirmed the effectiveness of each proposed module, demonstrating their individual contribution to enhancing the overall CFP mapping accuracy and boundary delineation quality.

The CNN-transformer fusion strategy designed in this study effectively integrates both the global and local information of CFPs, thereby significantly enhancing the discrimination between cropland and noncropland areas. Extensive migration experiments were conducted using Google Earth imagery across regions in Europe, Africa, Oceania, North America, and South America, where the proposed model achieved an OA above 94% and an F1-score exceeding 96%, fully demonstrating CTHBNet's strong cross-regional transferability and generalization capability. Furthermore, CTHBNet demonstrated strong performance on the FGFD, Ai4boundary, and Denmark benchmark datasets, confirming its wide adaptability and practical application potential in multisource, multiscale remote sensing scenarios. In summary, the proposed model can not only achieve high-precision CFP mapping but also exhibits strong cross-regional generalization, making it highly valuable for practical deployment and broader application.

## ACKNOWLEDGMENT

The Gaofen-2 (GF-2) data were downloaded from the China Centre for Resources Satellite Data and Application website (https://data.cresda.cn/#/2dMap). The authors are grateful to Mark Ackerley for the English language editing of the article.

## REFERENCES

[1] X. Wu et al., "The potential role of abandoned cropland for food security in China," *Resour., Conservation Recycling*, vol. 212, Jan. 2025, Art. no. 108004.

[2] J. Han, B. A. Bryan, and Y. Zhang, "Strategic cropland reserves can strengthen China's food security," *Nature Food*, vol. 5, no. 7, pp. 535–538, Jul. 2024.

[3] F. Xu et al., "Deep learning in cropland field identification: A review," *Comput. Electron. Agricult.*, vol. 222, Jul. 2024, Art. no. 109042.

[4] O. Deng et al., "Managing fragmented croplands for environmental and economic benefits in China," *Nature Food*, vol. 5, no. 3, pp. 230–240, Mar. 2024.

[5] A. Guo et al., "Cropland abandonment in China: Patterns, drivers, and implications for food security," *J. Cleaner Prod.*, vol. 418, Sep. 2023, Art. no. 138154.

[6] A. Owusu et al., "A framework for disaggregating remote-sensing cropland into rainfed and irrigated classes at continental scale," *Int. J. Appl. Earth Observ. Geoinformation*, vol. 126, Feb. 2024, Art. no. 103607.

[7] H. Wu et al., "CT-HiffNet: A contour-texture hierarchical feature fusion network for cropland field parcel extraction from high-resolution remote sensing images," *Comput. Electron. Agricult.*, vol. 239, Dec. 2025, Art. no. 111010.

[8] J. Li, K. Zheng, L. Gao, Z. Han, Z. Li, and J. Chanussot, "Enhanced deep image prior for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5504218.

[9] H. Yu, Y. Cui, S. Li, S. Kang, Z. Yao, and Z. Wei, "Estimation of the deep drainage for irrigated cropland based on satellite observations and deep neural networks," *Remote Sens. Environ.*, vol. 298, Dec. 2023, Art. no. 113819.

[10] X. Zhao, T. Wu, S. Wang, K. Liu, and J. Yang, "Cropland abandonment mapping at sub-pixel scales using crop phenological information and MODIS time-series images," *Comput. Electron. Agricult.*, vol. 208, May 2023, Art. no. 107763.

[11] J. Li, Y. Wei, T. Wei, and W. He, "A comprehensive deep-learning framework for fine-grained farmland mapping from high-resolution images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025.

[12] H. Zhao et al., "A large-scale VHR parcel dataset and a novel hierarchical semantic boundary-guided network for agricultural parcel delineation," *ISPRS J. Photogramm. Remote Sens.*, vol. 221, pp. 1–19, Mar. 2025.

[13] J. Li, K. Zheng, L. Gao, L. Ni, M. Huang, and J. Chanussot, "Model-informed multistage unsupervised network for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5516117.

[14] J. Li, K. Zheng, Z. Li, L. Gao, and X. Jia, "X-shaped interactive autoencoders with cross-modality mutual learning for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5518317.

[15] Y. Xie, S. Zheng, H. Wang, Y. Qiu, X. Lin, and Q. Shi, "Edge detection with direction guided postprocessing for farmland parcel extraction," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 3760–3770, May 2023.

[16] Z. Cai et al., "Improving agricultural field parcel delineation with a dual branch spatiotemporal fusion network by integrating multimodal satellite data," *ISPRS J. Photogramm. Remote Sens.*, vol. 205, pp. 34–49, Nov. 2023.

[17] B. Watkins and A. van Niekerk, "A comparison of object-based image analysis approaches for field boundary delineation using multi-temporal Sentinel-2 imagery," *Comput. Electron. Agricult.*, vol. 158, pp. 294–302, Mar. 2019.

[18] C. Wen et al., "An object-based genetic programming approach for cropland field extraction," *Remote Sens.*, vol. 14, no. 5, p. 1275, Mar. 2022.

[19] O. Vlachopoulos, B. Leblon, J. Wang, A. Haddadi, A. Larocque, and G. Patterson, "Delineation of bare soil field areas from unmanned aircraft system imagery with the mean shift unsupervised clustering and the random forest supervised classification," *Can. J. Remote Sens.*, vol. 46, no. 4, pp. 489–500, Jul. 2020.

[20] T. Cheng et al., "DESTIN: A new method for delineating the boundaries of crop fields by fusing spatial and temporal information from WorldView and planet satellite imagery," *Comput. Electron. Agricult.*, vol. 178, Nov. 2020, Art. no. 105787.

[21] M. Belgiu and O. Csillik, "Sentinel-2 cropland mapping using pixel-based and object-based time-weighted dynamic time warping analysis," *Remote Sens. Environ.*, vol. 204, pp. 509–523, Jan. 2018.

[22] J. Li, Y. Li, S. Zhang, and Y. Chen, "Image-point cloud embedding network for simultaneous image-based farmland instance extraction and point cloud-based semantic segmentation," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 136, Feb. 2025, Art. no. 104361.

[23] H. Wu, W. Mu, D. Zhong, Z. Du, H. Li, and C. Tao, "FarmSeg_VLM: A farmland remote sensing image segmentation method considering vision-language alignment," *ISPRS J. Photogramm. Remote Sens.*, vol. 225, pp. 423–439, Jul. 2025.

[24] Y. Pan, X. Wang, L. Zhang, and Y. Zhong, "E2EVAP: End-to-end vectorization of smallholder agricultural parcel boundaries from high-resolution remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 203, pp. 246–264, Sep. 2023.

[25] F. Waldner and F. I. Diakogiannis, "Deep learning on edge: Extracting field boundaries from satellite images with a convolutional neural network," *Remote Sens. Environ.*, vol. 245, Aug. 2020, Art. no. 111741.

[26] Z. Lv, S. Cheng, L. Xie, J. Li, and M. Zhao, "A graph contrastive learning network for change detection with heterogeneous remote sensing images," *Pattern Recognit.*, vol. 172, Apr. 2026, Art. no. 112394.

[27] Z. Zhang et al., "MATNet: Multiattention transformer network for cropland semantic segmentation in remote sensing images," *Int. J. Digit. Earth*, vol. 17, no. 1, Dec. 2024, Art. no. 2392845.

[28] C. Long et al., "Field patch extraction based on high-resolution imaging and U2-Net++ convolutional neural networks," *Remote Sens.*, vol. 15, no. 20, p. 4900, Oct. 2023.

[29] Z. Li et al., "Full convolution neural network combined with contextual feature representation for cropland extraction from high-resolution remote sensing images," *Remote Sens.*, vol. 14, no. 9, p. 2157, Apr. 2022.

[30] D. Zhang et al., "A generalized approach based on convolutional neural networks for large area cropland mapping at very high resolution," *Remote Sens. Environ.*, vol. 247, Sep. 2020, Art. no. 111912.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:08:14 UTC from IEEE Xplore. Restrictions apply.

[31] L. Xu, D. Ming, T. Du, Y. Chen, D. Dong, and C. Zhou, "Delineation of cultivated land parcels based on deep convolutional networks and geographical thematic scene division of remotely sensed images," *Comput. Electron. Agricult.*, vol. 192, Jan. 2022, Art. no. 106611.
[32] J. Xie et al., "Integrating random forest with boundary enhancement for mapping crop planting structure at the parcel level from remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 9934–9953, 2025.
[33] J. Wu, X. Ding, J. Wang, and J. Pan, "MAMNet: Lightweight multi-attention collaborative network for fine-grained cropland extraction from Gaofen-2 remote sensing imagery," *Agriculture*, vol. 15, no. 11, p. 1152, May 2025.
[34] J. Sheng et al., "HBRNet: Boundary enhancement segmentation network for cropland extraction in high-resolution remote sensing images," *Agriculture*, vol. 12, no. 8, p. 1284, Aug. 2022.
[35] B. Zhong et al., "Multi-swin mask transformer for instance segmentation of agricultural field extraction," *Remote Sens.*, vol. 15, no. 3, p. 549, Jan. 2023.
[36] Y. Xu, Z. Zhu, M. Guo, and Y. Huang, "Multiscale edge-guided network for accurate cultivated land parcel boundary extraction from remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4501020.
[37] Y. Liu, T. Zhang, Y. Huang, and F. Shi, "An edge-aware multitask network based on CNN and transformer backbone for farmland instance segmentation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 13765–13779, 2024.
[38] L. Xu et al., "Extraction of cropland field parcels with high resolution remote sensing using multi-task learning," *Eur. J. Remote Sens.*, vol. 56, no. 1, Dec. 2023, Art. no. 2181874.
[39] C. Persello, V. A. Tolpekin, J. R. Bergado, and R. A. de By, "Delineation of agricultural fields in smallholder farms from satellite images using fully convolutional networks and combinatorial grouping," *Remote Sens. Environ.*, vol. 231, Sep. 2019, Art. no. 111253.
[40] X. Liu, J. Zhang, Y. Duan, and J. Zhou, "CPVF: Vectorization of agricultural cultivation field parcels via a boundary–parcel multi-task learning network in ultra-high-resolution remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 226, pp. 267–299, Aug. 2025.
[41] S. Yan et al., "TSANet: A deep learning framework for the delineation of agricultural fields utilizing satellite image time series," *Comput. Electron. Agricult.*, vol. 220, May 2024, Art. no. 108902.
[42] L. Xia, J. Luo, Y. Sun, and H. Yang, "Deep extraction of cropland parcels from very high-resolution remotely sensed imagery," in *Proc. 7th Int. Conf. Agro-Geoinformatics (Agro-geoinformatics)*, Aug. 2018, pp. 1–5.
[43] J. Long, M. Li, X. Wang, and A. Stein, "Delineation of agricultural fields using multi-task BsiNet from high-resolution satellite images," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 112, Aug. 2022, Art. no. 102871.
[44] M. Li, J. Long, A. Stein, and X. Wang, "Using a semantic edge-aware multi-task neural network to delineate agricultural parcels from remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 200, pp. 24–40, Jun. 2023.
[45] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Las Vegas, NV, USA, Jun. 2016, pp. 770–778.
[46] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Montreal, QC, Canada, Oct. 2021, pp. 9992–10002.
[47] Z. Tian, T. He, C. Shen, and Y. Yan, "Decoders matter for semantic segmentation: Data-dependent decoding enables flexible feature aggregation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Long Beach, CA, USA, Jun. 2019, pp. 3121–3130.
[48] Z. Zhao, P. Wang, P. Zhang, and Z. Lv, "Sample augmentation and balance approach for improving classification performance with high-resolution remote sensed image," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025.
[49] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. Int. Conf. Learn. Represent.*, Dec. 2014, pp. 1–15.
[50] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. 18th Med. image Comput. Comput.-Assisted Intervent*, Nov. 2015, pp. 234–241.
[51] X. Xiao, S. Lian, Z. Luo, and S. Li, "Weighted res-UNet for high-quality retina vessel segmentation," in *Proc. 9th Int. Conf. Inf. Technol. Med. Educ. (ITME)*, Hangzhou, China, Oct. 2018, pp. 327–331.
[52] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder–decoder with atrous separable convolution for semantic image segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Oct. 2018, pp. 833–851.
[53] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 6230–6239.
[54] K. Sun, B. Xiao, D. Liu, and J. Wang, "Deep high-resolution representation learning for human pose estimation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 5686–5696.
[55] H. Cao et al., "Swin-UNet: UNet-like pure transformer for medical image segmentation," in *Proc. ECCV*, Oct. 2021, pp. 205–218.
[56] L. Wang et al., "UNetFormer: A UNet-like transformer for efficient semantic segmentation of remote sensing urban scene imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 190, pp. 196–214, Aug. 2022.
[57] H. Wu, P. Huang, M. Zhang, W. Tang, and X. Yu, "CMTFNet: CNN and multiscale transformer fusion network for remote-sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 2004612.
[58] B. Murugesan, K. Sarveswaran, S. M. Shankaranarayana, K. Ram, J. Joseph, and M. Sivaprakasam, "Psi-Net: Shape and boundary aware joint multi-task deep network for medical image segmentation," in *Proc. 41st Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, Jul. 2019, pp. 7223–7226.
[59] A. M. Kirillov et al., "Segment anything," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Jun. 2023, pp. 3992–4003.
[60] C. Lu, X. Zhang, K. Du, H. Xu, and G. Liu, "CTCFNet: CNN-transformer complementary and fusion network for high-resolution remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5408717.
[61] X. He, Y. Zhou, J. Zhao, D. Zhang, R. Yao, and Y. Xue, "Swin transformer embedding UNet for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4408715.
[62] H. Wang, X. Chen, T. Zhang, Z. Xu, and J. Li, "CCTNet: Coupled CNN and transformer network for crop segmentation of remote sensing images," *Remote Sens.*, vol. 14, no. 9, p. 1956, Apr. 2022.
[63] R. D'Andrimont et al., "AI4Boundaries: An open AI-ready dataset to map field boundaries with Sentinel-2 and aerial photography," *Earth Syst. Sci. Data*, vol. 15, no. 1, pp. 317–329, Jan. 2023.



**Junyang Xie** received the B.S. degree in geographic information science from Chengdu University of Information Technology, Chengdu, China, in 2019, and the M.S. degree in cartography and GIS from Chengdu University of Technology, Chengdu, in 2022. He is currently pursuing the Ph.D. degree in cartography and GIS with Central China Normal University, Wuhan, China.

His research interests include agricultural remote sensing and deep-learning algorithms.



**Hao Wu** (Senior Member, IEEE) received the B.S. degree in survey engineering and the M.S. degree in geodesy and survey engineering degrees from the School of Geodesy and Geomatics, Wuhan University, Wuhan, China, in 2000 and 2003, respectively, and the Ph.D. degree in cartography and GIS from the State Key Laboratory of Information Engineering in Surveying, Mapping, and Remote Sensing, Wuhan University, in 2007.

He is currently a Professor with the College of Urban and Environmental Sciences, Central China Normal University, Wuhan. His research interests include earth observation science and its applications.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:08:14 UTC from IEEE Xplore. Restrictions apply.



**Wenbin Wu** received the B.S. degree in geography from Central China Normal University, Wuhan, Hubei, China, in 1998, the M.S. degree in environmental engineering from the Chinese Academy of Agricultural Sciences, Beijing, China, in 2005, and the Ph.D. degree in agricultural spatial informatics from the University of Tokyo, Tokyo, Japan, in 2008.

He is a Research Fellow with the Institute of Agricultural Resources and Regional Planning, Chinese Academy of Agricultural Sciences. He is mainly engaged in the field of agricultural land systems.

**Liang Hong**, is currently with Hubei Provincial Geographic National Conditions Monitoring Center, Wuhan, China. His research interests include remote sensing of natural resources.

**Lihua He**, is currently with Hubei Provincial Geographic National Conditions Monitoring Center, Wuhan, China. Her research interests include remote sensing of natural resources.



**Qiangyi Yu** received the B.S. degree in land resources management from Hunan Agricultural University, Changsha, China, in 2007, and the M.S. degree in agricultural regional development and the Ph.D. degree in agricultural resources and environment from the Chinese Academy of Agricultural Sciences, Beijing, China, in 2010 and 2013, respectively.

He is a Research Fellow with the Institute of Agricultural Resources and Regional Planning, Chinese Academy of Agricultural Sciences, and a Leader Assistant of the Agricultural Land System Group. His research interests include advancing the manifestations of agricultural land system by considering crop allocation, farm management, and all-inclusive human activities that are relevant to the state of cropland.



**Lanfa Liu** received the Ph.D. degree from the Technical University of Dresden, Dresden, Germany, in 2018.

He is currently an Associate Professor with the College of Urban and Environmental Sciences, Central China Normal University, Wuhan, China. He is also with the State Key Laboratory of Lunar and Planetary Sciences, Macau University of Science and Technology, Macau, China. His research interests include geoinformatics and remote sensing data processing and applications.



**Anqi Lin** received the B.S. degree in GIS from Wuhan University of Technology, Wuhan, China, in 2014, the M.S. degree in GIS from the University of Pittsburgh, Pittsburgh, PA, USA, in 2016, and the Ph.D. degree in geography from Central China Normal University, Wuhan, China, in 2021.

She is currently an Assistant Professor with the College of Urban and Environmental Sciences, Central China Normal University. Her research interests include geographic information science, remote sensing, and spatial big data.

**Jaturong Som-ard**, is working on machine learning and deep learning for agriculture, land use mapping, and natural resources management. Moreover, his research interests are crop type and land cover classification, spatial analysis, and weather effect monitoring using EO data and remote sensing techniques.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:08:14 UTC from IEEE Xplore. Restrictions apply.